Representative Action Selection for Large Action Space Bandit Families

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Abstract

We study the problem of selecting a subset from a large action space shared by a family of bandits, with the goal of achieving performance nearly matching that of using the full action space. Indeed, in many natural situations, while the nominal set of actions may be large, there also exist significant correlations between the rewards of different actions. In this paper we propose an algorithm that can significantly reduce the action space when such correlations are present, without the need to a-priori know the correlation structure. We provide theoretical guarantees on the performance of the algorithm and demonstrate its practical effectiveness through empirical comparisons with Thompson Sampling and Upper Confidence Bound methods.

1. Introduction

In this paper we study a family of bandits that share a common but very large action space. Our aim is to understand whether it is possible—and how—to select a smaller set of representative actions that performs nearly as well as the full action space across all bandit instances. As an example, consider a pharmacy preparing its inventory for the upcoming season. The set of available drugs will correspond to actions, and can be extremely large. Each patient will be represented by a bandit instance, i.e., a unique mapping from drugs to health outcomes based on their physiology and history. However, when multiple drugs contain similar ingredients, their effects on patients are likely to be similar. Therefore, we can retain just one or a few representative drugs from such a group without meaningfully reducing coverage or effectiveness.

In the classical multi-armed bandit problem (MAB) (Lai & Robbins, 1985; Lattimore & Szepesvári, 2020), the objective is to maximize cumulative reward over a time horizon T for a *single, fixed bandit*. This is achieved by exploiting a single optimal action after exploration over the action space. In contrast, our objective is to select an action subset that likely contains near-optimal actions jointly for a *family of bandits*. This problem is a large-scale combinatorial optimization under uncertainty (Van Hentenryck & Bent, 2006; Li & Yuan, 2013).It arises in settings with too many options to evaluate exhaustively— such as route planning (Feng et al., 2025) and inventory management (Namir et al., 2022)—due to computational or time constraints.

Consider a Bayesian multi armed bandits setting, (Slivkins et al., 2019; Lattimore & Szepesvári, 2020): Let $\mathcal{F}$ be a family of bandits which share an action space $\mathcal{A}_{\text{full}}$, and let $\mathcal{P}$ be a probability distribution over $\mathcal{F}$. For a bandit instance $\theta \in \mathcal{F}$, let $\mu_a = \mu_a(\theta)$ denote the expected reward of an action $a \in \mathcal{A}_{\text{full}}$ given θ. Each μ_a is then a random variable as θ is sampled from $\mathcal{P}$, and the collection $\{\mu_a\}_{a\in\mathcal{A}_{\text{full}}}$ constitutes a stochastic process indexed by the action space.

In this paper we consider the similarity structure on the actions $\mathcal{A}_{\text{full}}$ that is induced by the correlations of the variables μ_a. In this setting, the actions a and a' will be considered similar if the rewards μ_a and μ'_a are similar for most bandits sampled from $\mathcal{P}$. Note that this notion does not depend on the nominal properties of the set $\mathcal{A}_{\text{full}}$, such as the dimension in which $\mathcal{A}_{\text{full}}$ happens to be represented, but depends only on the statistical properties of the family of bandits $\mathcal{F}$ and the distribution $\mathcal{P}$. This provides a robust practical modeling framework which in turn may be used as the basis for the action subset selection — in a cluster of highly positively correlated actions, one may retain just one, without considerably reducing the performance.

1.1. Reward Approximation

Now, suppose that the decision-maker has access to a certain subset of actions, $\mathcal{A} \subset \mathcal{A}_{\text{full}}$, rather than to the full set $\mathcal{A}_{\text{full}}$. For a bandit instance θ, define the regret of being restricted to subset $\mathcal{A}$ as the gap between the best outcome achievable with the full action space and the best outcome within $\mathcal{A}$:

$$\text{Regret}(\theta) := \max_{a\in\mathcal{A}_{\text{full}}} \mu_a(\theta) - \max_{a'\in\mathcal{A}} \mu_{a'}(\theta), \qquad (1)$$

Our objective then is to identify a small subset $\mathcal{A}$ that minimizes the expected regret $\mathbb{E}_{\theta\sim\mathcal{P}}[\text{Regret}]$, where the expectation is taken over all possible bandit instances.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

This objective may be motivated in several ways: First, as discussed above, in problems such as inventory management it may be of natural interest to find a representative subset of actions, due to the intrinsic difficulties associated with maintaining the full set of actions.

Another possible reason would be to reduce the exploration phase when encountering a new bandit instance, especially when the time horizon T is small. Indeed, the classic Bayesian bandit regret, (Agrawal & Goyal, 2012), generally scales as $\Omega(\sqrt{|\mathcal{A}_{\text{full}}|\,T})$, and may become prohibitively large for large $|\mathcal{A}_{\text{full}}|$. On the other hand, if a subset $\mathcal{A}$ is carefully chosen, the resulting Bayesian bandit regret can be significantly lower. To see this, for a given $\mathcal{A} \subset \mathcal{A}_{\text{full}}$ we can decompose the bandit regret as follows:

$$\begin{aligned}
&\text{BayesianBanditRegret} := \mathbb{E}\sum_{t=1}^{T}\left[\max_{a\in\mathcal{A}_{\text{full}}}\mu_a(\theta) - \mu_{A_t}(\theta)\right] \\
&= \mathbb{E}\sum_{t=1}^{T}\left[\max_{a\in\mathcal{A}}\mu_a(\theta) - \mu_{A_t} + \max_{a\in\mathcal{A}_{\text{full}}}\mu_a(\theta) - \max_{a\in\mathcal{A}}\mu_a(\theta)\right] \\
&\leq C\sqrt{|\mathcal{A}|\cdot T\log T} + T\cdot\mathbb{E}_\theta[\text{Regret}]. \qquad (2)
\end{aligned}$$

In the first equality, $A_t \in \mathcal{A}$ denotes the action selected in round t by a policy such as Thompson Sampling (TS) (Agrawal & Goyal, 2012). The expectation is taken over the randomness in the distribution of bandit instances and in actions selected by the policy. The inequality (2) then follows from the well-known regret bound for TS (Agrawal & Goyal, 2012; Lattimore & Szepesvári, 2020), and the definition of regret in (1). Thus, we can switch from a large regret $\sqrt{|\mathcal{A}_{\text{full}}|\cdot T\log T}$, to a possibly much smaller one—$\sqrt{|\mathcal{A}|\cdot T\log T}$, although at a constant (per time step) price of $\mathbb{E}_\theta[\text{Regret}]$. Practically, this means that by restricting to $\mathcal{A}$, the algorithm will spend much less time exploring similar options, and the above tradeoff may be especially worthwhile for small T, i.e., when rapid adaptation to a new bandit instance is desired.

A further motivation is to reduce the computational burden that persists even in structured bandits; see Remark 4.9 for a discussion of linear and Gaussian process (GP) bandits.

1.2. Subset Selection Algorithm

We now introduce the subset selection algorithm itself, Algorithm 1. Conceptually, the algorithm is simple: Sample a bandit instance θ from a distribution $\mathcal{P}$ and solve it using an oracle (or practically, any of the standard bandit algorithms, such as UCB or TS), to find the optimal action $a^* = a^*(\theta)$. Add the action to the set or representation actions $\mathcal{A}$. Repeat the process K times.

In the rest of the paper we show that the action subset generated by this simple algorithm will be diverse enough to appropriately cover $\mathcal{A}_{\text{full}}$ by $\mathcal{A}$. Our bounds will depend on

Algorithm 1 Epsilon Net Algorithm

Input: Action space $\mathcal{A}_{\text{full}}$, Sample size K.
Output: A subset of actions $\mathcal{A}$.
$\mathcal{A} \leftarrow \emptyset$
for $1, \ldots, K$ **do**
  Sample a bandit instance θ from a distribution $\mathcal{P}$ (unknown to the learner).
  Find optimal action $a^*(\theta) := \arg\max_{a\in\mathcal{A}_{\text{full}}}\mu_a(\theta)$.
  $\mathcal{A} \leftarrow \mathcal{A} \cup \{a^*\}$.
end for

the metric structure of the set $\mathcal{A}_{\text{full}}$ under the L_2 metric, and on the number of samples K. Note that Algorithm 1 does not require the knowledge of the metric, only the ability to sample bandit instances at random, and to solve them.

Our main contributions are organized as follows:

- **Problem Formalization and An Algorithm** Although we believe the subset selection problem is natural and of practical interest, to the best of our knowledge it has not been studied in the literature. In this paper we provide the first formalization of the problem, in the framework of the statistical similarities between actions. We also introduce a simple but effective subset selection algorithm, that does not require apriori knowledge of the correlations.

- **Regret Analysis.** When the action sets exhibits sufficient correlation structure (ex. clustering or a low dimensional manifold structure), we provide upper and lower bounds on the expected error of the algorithm, i.e. the regret (1) (Section 4). The analysis is done under the standard assumption that the reward process $\{\mu_a\}_{a\in\mathcal{A}_{\text{full}}}$ is Gaussian, (Srinivas et al., 2009; Garnett, 2023) and shows the basic relations between the induced geometry of the action set and the expected rewards. Our arguments can also be applied virtually with no changes to the sub-Gaussian processes, and this extension is carried out in Supplementary I. An extension of these arguments to other types of tail behavior may be implemented using standard techniques.

- **Generalization and Empirical Validation.** We extend the analysis to settings where outcome functions lie in a reproducing kernel Hilbert space (RKHS). We empirically evaluate the subset selection algorithm on a number of problems (Section 5). Although there are no existing algorithms that could serve as direct performance comparison benchmarks, one could naively apply standard bandit algorithms, such as TS and UCB, via the notions of "super-arms". As expected, since these methods are not sensitive to the underlying geometry of the subsets, we show that their performance is much weaker than that of our approach.

2. Related Work

For a general overview of multi-armed bandits (MAB), see (Lattimore & Szepesvári, 2020; Slivkins et al., 2019). Background on optimal action identification—which addresses the sample complexity of the inner optimization step in Algorithm 1—is provided in Supplementary A, along with regret bounds for structured (linear/GP) bandits.

While regret in general Bayesian bandits scales with $|\mathcal{A}_{\text{full}}|$, linear/GP bandits achieve $|\mathcal{A}_{\text{full}}|$-independent regret that depends on structural complexity (e.g., feature dimension or information gain). Their computational complexity, however, typically remains $|\mathcal{A}_{\text{full}}|$-dependent. Further discussion of computational aspects appears in Remark 4.9.

Before discussing two related approaches that select a fixed-size subset $|\mathcal{A}| = K$–a choice often unclear in practice and requiring algorithm restarts–we highlight a key advantage of our method: it can adapt the subset cardinality on the fly.

Top-K Action Identification aims to identify the K actions with the highest expected outcomes using as few samples as possible (Kalyanakrishnan et al., 2012; Gabillon et al., 2012; Kaufmann et al., 2016; Chen et al., 2017). This line of work assumes that all actions are independent and have distinct expected values, making its methods inapplicable to our framework. If one were to apply these methods regardless, the most reasonable approach, in our view, would be to treat the family of bandit instances as a super-bandit, where each bandit instance corresponds to a round, and the expected payoff of an action in that round is given by μ_a. In this setting, top-K identification would refer to selecting K actions with the highest expected payoffs $\mathbb{E}[\mu_a]$. In contrast, our framework considers that the expected outcome of each action, averaged over the distribution of bandits, may be the same—i.e., $\mathbb{E}[\mu_a]=c$ for all $a\in\mathcal{A}_{\text{full}}$, where c is a constant. Further, even if $\mathbb{E}[\mu_a]$ varies across actions, ignoring correlations can be fatal in our framework:

Example 2.1. Consider three actions: $a_1 = [1, 0]$, $a_2 = [0.9, 0.1]$, and $a_3 = [-0.1, 1]$, and suppose bandits are sampled uniformly from $\theta_1 = [1, 0]$ and $\theta_2 = [0, 1]$. Then,

$$\mathbb{E}\langle a_1, \theta\rangle = \mathbb{E}\langle a_2, \theta\rangle = 0.5, \quad \mathbb{E}\langle a_3, \theta\rangle = 0.45.$$

So under the Best-2-Action perspective, a_1 and a_2 would be selected. However, this is suboptimal in our framework, since μ_{a_1} and μ_{a_2} are positively correlated:

$$\mathbb{E}\max_{a\in\{a_1,a_2\}}\langle a, \theta\rangle = 0.55, \quad \mathbb{E}\max_{a\in\{a_1,a_3\}}\langle a, \theta\rangle = 1.$$

Our algorithm, if run until it selects two distinct actions, would output a_1 and a_3—the true optimal.

Combinatorial Bandits considers that the decision maker selects K of base arms from $\mathcal{A}_{\text{full}}$ in each round, forming a super-arm $\mathcal{A}$, with $|\mathcal{A}| = K$. Popular methods include CUCB (Chen et al., 2016), CTS (Wang & Chen, 2018). We include this line of work as an empirical baseline, though it is not directly applicable to our framework for two reasons: (1) It assumes that the expected outcome of a super-arm depends only on the expected outcomes of its individual base arms, or imposes a stricter monotonicity condition. In our case, even though $\mathbb{E}[\mu_a] = c$ for all $a \in \mathcal{A}_{\text{full}}$, super-arm expected outcomes $\mathbb{E}[\max_{a\in\mathcal{A}} \mu_a]$ can differ significantly due to correlations among actions. (2) It assumes independence across base arms, whereas we explicitly model correlations. Ignoring these correlations misses the core challenge—an issue illustrated in Example 2.1.

3. Subset Selection Algorithm

In this section, we introduce the formal problem setup and the technical tools used in the analysis.

The optimal action in a bandit instance θ is defined as

$$a^*(\theta) := \arg\max_{a\in\mathcal{A}_{\text{full}}} \mu_a(\theta). \tag{3}$$

Thus a^* is a random variable taking values in the set $\mathcal{A}_{\text{full}}$. We denote by q the induced probability distribution of a^* on $\mathcal{A}_{\text{full}}$. Specifically, for any subset $r \subset \mathcal{A}_{\text{full}}$, we have

$$q(r) := \mathbb{P}\left(a^*(\theta) \in r\right) = \mathbb{E}_{\theta\sim\mathcal{P}}\mathbb{1}_{\{a^*(\theta)\in r\}}. \tag{4}$$

In particular, note that the samples a^* in lines 5&6 of Algorithm 1 may be viewed as i.i.d samples from q. The distribution q will be named as the *importance measure* q.

3.1. Gaussian Processes

As discussed earlier, in our analysis we consider the setting where the reward expectations $\{\mu_a\}_{a\in\mathcal{A}_{\text{full}}}$ form a Gaussian process. An extension of our main bound to sub Gaussian processes is given in Supplementary I.

Let $\|\cdot\|_2$ denote the standard Euclidean norm on $\mathbb{R}^n$, and for a set $S \subset \mathbb{R}^n$ define the diameter of S as $\text{diam}(S) := \sup_{a,b\in S} \|a - b\|_2$.

To simplify the notation, we will consider the canonical representation of the Gaussian process: That is, we assume that $\mathcal{A}_{\text{full}} \subset \mathbb{R}^n$ for some dimension $n > 0$, and that

$$\mu_a(\theta) := \langle a, \theta\rangle, \quad \forall a \in \mathcal{A}_{\text{full}}, \theta \sim \mathcal{N}(0, I). \tag{5}$$

Here, $\langle\cdot,\cdot\rangle$ is the standard inner product on $\mathbb{R}^n$, and $\mathcal{N}(0, I)$ is the standard Gaussian distribution on $\mathbb{R}^n$. Thus, in terms of the notation in Section 1, the family of bandits $\mathcal{F}$ may be identified as $\mathbb{R}^n$, and the distribution $\mathcal{P}$ as the standard Gaussian. Clearly, the family of variables $\{\mu_a\}_{a\in\mathcal{A}_{\text{full}}}$ given by (5) is a Gaussian process. Conversely, we note that any finite dimensional Gaussian process can be represented as (5), and any infinite dimensional Gaussian process[1] can

[1] Under very mild technical assumptions

be arbitrarily approximated by (5), in a sufficiently high dimension n. See (Vershynin, 2018, Chapter 7). Thus, we may assume the form (5) without loss of generality.

Moreover, crucially, the dimension n will not be important in our bounds, and can be large. On the other hand, however, it may be also of interest to note that when n is fixed, the setup (5) is equivalent to the Bayesian *linear bandit* setup, (Lattimore & Szepesvári, 2020) (although we solve a different problem). The relation with linear bandits will be discussed further in Remark 4.9.

We define the L_2 distance induced on the $\mathcal{A}_{\text{full}}$ by the process to be the distance

$$\|\mu_a - \mu_{a'}\|_{L_2}^2 := \mathbb{E}_\theta \left|\langle a, \theta\rangle - \langle a', \theta\rangle\right|^2 . \tag{6}$$

Note that under the definition (5), the L_2 distance and the standard Euclidean distance in $\mathbb{R}^n$ coincide, $\|\mu_a - \mu_{a'}\|_{L_2}^2 = \|a - a'\|_2^2$ for all $a, a' \in \mathcal{A}_{\text{full}}$.

3.2. Epsilon Nets

There will be two senses in which we would be interested in approximating $\mathcal{A}_{\text{full}}$: First, for every $a \in \mathcal{A}_{\text{full}}$ we would like to have $a' \in \mathcal{A}$ such that a' is close to a in the distance (6) (or, equivalently, in $\|\cdot\|_2$). Second, in order for the subset selection algorithm to able to discover the points a', we need an additional, probabilistic notion of approximation.

The following two notions of an ϵ-net, the *geometric* and the *measure theoretic* nets, formalize these approximations. We refer to (Vershynin, 2018, Chapter 4) for additional details on geometric nets, and to (Matousek, 2013, Chapter 10) for measure theoretic nets.

Definition 3.1. A subset $\mathcal{A} \subseteq \mathcal{A}_{\text{full}}$ is called a *geometric ϵ-net* if, for all $a \in \mathcal{A}_{\text{full}}$, there exists $a' \in \mathcal{A}$ such that

$$\|a - a'\|_2 < \epsilon.$$

As discussed earlier, geometric ϵ-nets will be used as approximations of the set $\mathcal{A}_{\text{full}}$. While we'll assume that good approximating nets exist in the data, these nets will never need to be explicitly constructed. Instead, part of such representative points will be implicitly chosen by the subset selection algorithm.

The *ϵ-covering number* of the set $\mathcal{A}_{\text{full}}$, denoted $N(\mathcal{A}_{\text{full}}, \epsilon)$, is the size of the smallest ϵ-net of $\mathcal{A}_{\text{full}}$:

$$N(\mathcal{A}_{\text{full}}, \epsilon) = \min \left\{|\mathcal{A}'| \;\mid\; \mathcal{A}' \text{ is an } \epsilon\text{-net of } \mathcal{A}_{\text{full}}\right\} . \tag{7}$$

Now, Let $\mathcal{R}$ be a finite partition of $\mathcal{A}_{\text{full}}$ into disjoint clusters r, such that $\cup_{r\in\mathcal{R}}\ r = \mathcal{A}_{\text{full}}$. Let q be a probability distribution on $\mathcal{A}_{\text{full}}$, which therefore assigns a weight $q(r)$ to each cluster $r \in \mathcal{R}$.

Definition 3.2. A subset $\mathcal{A} \subseteq \mathcal{A}_{\text{full}}$ is called a *measure-theoretic ϵ-net* with respect to measure q and partition $\mathcal{R}$, if, for any cluster $r \in \mathcal{R}$, we have:

$$r \cap \mathcal{A} \neq \emptyset \quad \text{whenever} \quad q(r) > \epsilon.$$

The measure theoretic ϵ-net ensures that in a given partition $\mathcal{R}$, either $\mathcal{A}$ intersects a cluster r, or the cluster r is small in terms of the distribution q. Roughly speaking, this will be used to show that if the subset selection algorithm doesn't find a cluster r of actions, then the cluster's contribution to the total expected regret is small and can be ignored.

An important property of random sampling is that with high probability it will produce a measure theoretic ϵ-net for *any* fixed partition of the support of the distribution. Specifically, we have the following result:

Lemma 3.3. *Fix $\epsilon > 0$. Let $\mathcal{R}$ be a partition of $\mathcal{A}_{\text{full}}$, and let q be an importance measure given by (4). Let $\mathcal{A}$ be the output of Algorithm 1 after K samples. Then, with probability at least $1 - \frac{1}{\epsilon}\exp(-K\epsilon)$, it holds that for any cluster $r \in \mathcal{R}$,*

$$r \cap \mathcal{A} \neq \emptyset \quad \textit{whenever} \quad q(r) > \epsilon.$$

This result is a special case of Theorem 10.2.4 in (Matousek, 2013), which in turn follows the analysis in (Haussler & Welzl, 1986). The algorithm in (Haussler & Welzl, 1986) is called the ϵ-net Algorithm, and we have retained that name. The full proof of the Lemma is given in Supplementary C.

4. Regret Analysis

In this Section we introduce and discuss the regret bounds for subset selection Algorithm 1. Our main result is Theorem 4.4, which bounds the regret in terms of the covering numbers of the set $\mathcal{A}_{\text{full}}$. Typically, sets which exhibit clustering, or sets with low dimensional manifold structure, would have low covering numbers, and we expect such sets to commonly occur in practical problems (the manifold hypothesis, (Bengio et al., 2013)). Roughly speaking, suppose that the set $\mathcal{A}_{\text{full}}$ can be represented as a union of a few small clusters, as illustrated in Figure 1. The main term in regret bound bound in Theorem 4.4 is related to the complexity of each individual cluster, and typically would controlled by its diameter ϵ. Additional terms involve a logarithmic dependence on the total number of clusters (times ϵ), and an error term which accounts for the probability of Algorithm 1 to find the relevant clusters. This term decays exponentially with K.

Perhaps the most central technical notion in our analysis is the notion of *partition* $\mathcal{R} = \{r_\ell\}_{\ell\leq m}$, such that $\mathcal{A}_{\text{full}} = \cup_{l\leq m} r_\ell$, and we will typically be interested in partitions where each subset r_ℓ also have a relatively small diameter

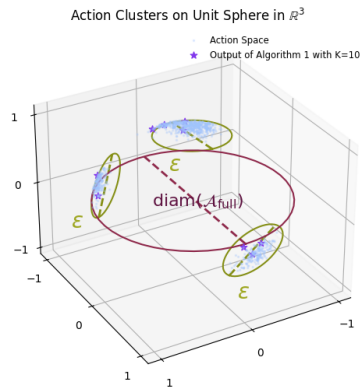

Figure 1. Illustration of Theorem 4.4. The full action space (blue dots) lies on the unit sphere in $\mathbb{R}^3$ and can be partitioned into three clusters. The regret bound established in the theorem depends not on the diameter of the entire action space—indicated by the red dashed line—but rather on the diameters of the individual clusters, represented by the green dashed lines, each of which is significantly smaller.

ϵ. There are two main ways in which such partitions may be present in $\mathcal{A}_{\text{full}}$. First, some realistic datasets may be naturally clustered, as depicted in Figure 1. Typically, in such situations there will be a characteristic scale ϵ, such that there are few clusters with diameter ϵ, which are well separated. However, inside each cluster, on scales smaller then ϵ, there may be no additional structure, i.e. the clusters themselves would look dense and high dimensional. In such situations, we still may benefit from a regret of order ϵ. The second scenario is when the whole set $\mathcal{A}_{\text{full}}$ is a low dimensional manifold, typically in a high dimensional space. In that case, for any scale ϵ we construct relatively small ϵ-nets, which can be used to derive appropriate partitions.

Section 4.1 introduces bounds on the regret in terms of the partitions and their diameters. These are the bounds that one could obtain for a fixed a-priori *known* partitions. In Section 4.2 we obtain bounds for the subset selected by Algorithm 1, which finds (an approximation of) the reference points for the best partition available in $\mathcal{A}_{\text{full}}$, without knowing this partition. Theorem 4.4 provides bounds in terms of a general partition given scale ϵ, while in Theorem 4.10, by taking ϵ to 0 we provide worst case diminishing bounds.

4.1. Partitions and Reference Set Bounds

For a point $a \in \mathbb{R}^n$ and scalar $\epsilon \geq 0$ denote by $B(a, \epsilon) := \{a' \in \mathbb{R}^n \mid \|a - a'\|_2 \leq \epsilon\}$ the closed Euclidean ball of radius ϵ around a.

Definition 4.1 (ϵ-reference sets)**.** Consider a partition $\mathcal{R} := \{r_\ell\}_{\ell \leq m}$ of $\mathcal{A}_{\text{full}}$. For an $\epsilon > 0$, a set $\mathcal{A}^\epsilon := \{a_1, \ldots, a_m\}$ is an *ϵ-reference set* for $\mathcal{R}$ if for every $\ell \leq m$, $r_\ell \subseteq B(a_\ell, \epsilon)$. The point a_ℓ is called the *reference point* of the subset r_ℓ.

Note that the ϵ-reference set is in particular a (geometric) ϵ-net of $\mathcal{A}_{\text{full}}$. However, for our purposes it will be more convenient to discuss partitions, which we do with the notion of reference sets. Note also that given a partition $\mathcal{R}$, one can always find a reference set with $\epsilon = \max_{r \in \mathcal{R}} \operatorname{diam}(r)$.

Theorem 4.2 (Regret bounds of ϵ-reference subsets)**.** *Let $\mathcal{R}$ be a partition of $\mathcal{A}_{\text{full}}$ and let $\mathcal{A}^\epsilon$ be an ϵ-reference subset for $\mathcal{R}$. There is an absolute constant $C > 0$, such that*

$$\mathbb{E}_\theta[\text{Regret}] \leq \max_{\ell \leq m} \mathbb{E}_\theta \left[\max_{a \in r_\ell} (\mu_a - \mu_{a_\ell}) \right] + C\epsilon\sqrt{\log m}.$$

Proof sketch. For each cluster $\ell \leq m$, define a simple GP $\{Z_a\}_{a \in r_\ell}$ where $Z_a := \mu_a - \mu_{a_\ell}$. Define a non-negative random variable $Y_\ell := \sup_{a \in r_\ell} Z_a$. When $a^*(\theta) \in r_\ell$, regret is upper bounded by Y_ℓ. Thus, for any bandit θ, the regret is bounded between $\min_{\ell \leq m} Y_\ell$ and $\max_{\ell \leq m} Y_\ell$. Finally, the expectations $\mathbb{E}\left[\min_{\ell \leq m} Y_\ell\right]$ and $\mathbb{E}\left[\max_{\ell \leq m} Y_\ell\right]$ can be bounded via appropriate concentration arguments. □

We give the corresponding lower bound. We define the outer radius induced by an action subset $\mathcal{A}$ as

$$\Delta(\mathcal{A}) := \max_{a \in \mathcal{A}} \sup_{a' \in \mathcal{A}_{\text{full}}} \|a - a'\|_2,$$

which satisfies $\Delta(\mathcal{A}) \leq \operatorname{diam}(\mathcal{A}_{\text{full}})$.

Theorem 4.3 (Lower bound of ϵ-reference subsets)**.** *Let $\mathcal{R}$ be a partition of $\mathcal{A}_{\text{full}}$ and let $\mathcal{A}^\epsilon$ be an ϵ-reference subset for $\mathcal{R}$. There is an absolute constant $C > 0$, such that*

$$\mathbb{E}_\theta[\text{Regret}] \geq \max_{\ell \leq m} \mathbb{E} \left[\max_{a \in r_\ell} (\mu_a - \mu_{a_\ell}) \right] - C\, \Delta(\mathcal{A}^\epsilon)\, \sqrt{\log m}.$$

Proof sketch. Define $Y_{\ell,\ell'} := \sup_{a \in r_\ell \cup \{a_{\ell'}\}} (\mu_a - \mu_{a_{\ell'}})$. When $a^*(\theta) \in r_\ell$, we have an interesting property that

$$\text{Regret} = \min_{\ell'} Y_{\ell,\ell'} = \min_{\ell'} \max_{\ell''} Y_{\ell'',\ell'}.$$

Decomposing $Y_{\ell,\ell'}$ into its mean and deviation, the fluctuation term is controlled uniformly via Gaussian concentration, yielding a $\Delta(\mathcal{A}^\epsilon)\sqrt{\log m^2}$ penalty. Finally, a min–max argument shows

$$\min_{\ell'} \max_{\ell} \mathbb{E} Y_{\ell,\ell'} \geq \max_{\ell} \mathbb{E} \left[\max_{a \in r_\ell} (\mu_a - \mu_{a_\ell}) \right],$$

which gives the result. □

4.2. Regret Bounds for the Algorithm

As discussed above, Theorem 4.4 provides regret bounds for the regret of the set $\mathcal{A}$ selected by Algorithm 1.

To get an intuition about the argument, suppose that $\mathcal{A}_{\text{full}}$ admits a good partition $\mathcal{R}$ (i.e. relatively few clusters, of

small diameter ϵ). We then show that for any fixed partition $\mathcal{R}$, samples in Algorithm 1 will hit most of the clusters of that partition, and can consequently be used as reference set points in Theorem 4.2. Indeed, by Lemma 3.3, either Algorithm 1 will hit the cluster, or the probability of that cluster under q is small, in which case its contribution to regret will be small too. This gives rise to the additional error term in Theorem 4.4 compared to Theorem 4.2, but this term decays exponentially with K.

Theorem 4.4. *Let $\mathcal{R}$ be a partition of $\mathcal{A}_{\text{full}}$ and set $\epsilon = \max_{r\in\mathcal{R}} diam(r)$. Let $\mathcal{A}$ be the output of Algorithm 1 after K iterations. Then*

$$\mathbb{E}_{\theta,\mathcal{A}}[\text{Regret}] \leq \max_{\ell\leq m} \mathbb{E}_\theta\left[\max_{a\in r_\ell}(\mu_a - \mu_{a_\ell})\right] + C\epsilon\sqrt{\log m} + \left(\mathbb{E}_q\left[(1-q(r))^{2K}\right]\cdot \mathbb{E}_\theta\left[\max_{a\in\mathcal{A}_{\text{full}}} \mu_a^2\right]\right)^{1/2},$$

where $C > 0$ is an absolute constant.

Note that the expectation $\mathbb{E}_{\theta,\mathcal{A}}$ in this result is taken with respect to both the algorithm's randomness (i.e. the sampled set $\mathcal{A}$) and the distribution over θ.

In the following remarks we discuss the different error terms appearing in Theorem 4.4.

Remark 4.5 (The Cluster Complexity Term). The term $\mathbb{E}_\theta\left[\max_{a\in r_\ell}\mu_a - \mu_{a_\ell}\right]$ in Theorem 4.4 measures the size of the cluster, and is also known as the *Gaussian width* of r_ℓ (see (Vershynin, 2018), Chapter 7). There are two standard ways in which we can bound this term: First, for any finite r_ℓ we have

$$\mathbb{E}_\theta\left[\max_{a\in r_\ell}\mu_a - \mu_{a_\ell}\right] \leq \frac{\text{diam}(r_\ell)}{2}\sqrt{\log|r_\ell|} \leq \frac{\epsilon}{2}\sqrt{\log|\mathcal{A}_{\text{full}}|}, \tag{8}$$

thus yielding a scaling with the diameter and a square-root-logarithmic dependence on the size of the actions set. See Supplementary K additional details on (8). In addition, we have the bound

$$\mathbb{E}_\theta\left[\max_{a\in r_\ell}\mu_a - \mu_{a_\ell}\right] \leq \text{diam}(r_\ell)\frac{\sqrt{n}}{2}, \tag{9}$$

where n is the ambient dimension of the action set, i.e. $\mathcal{A}_{\text{full}} \subset \mathbb{R}^n$ (see (Vershynin, 2018), Chapter 7). This bound is independent of the cardinality of the action set and may be applied if n is known to be small. However, as discussed in Section 3.1, we do not assume that n is small. We use both bounds in Theorem 4.10 below.

Remark 4.6 (Sampling Correction Term). The sampling correction term $\mathbb{E}_q\left[(1-q(r))^{2K}\right]$ in Theorem 4.4 decays exponentially with K. Here we analyze this decay in more detail. Write

$$\mathbb{E}_q\left[(1-q(r))^{2K}\right] = \sum_r q(r)(1-q(r))^{2K} \leq \sum_r q(r)e^{-2Kq(r)},$$

where we have used $1 - x \leq e^{-x}$ for $x \in [0,1]$. For each cluster $r \in \mathcal{R}$, the summand $q(r)e^{-2Kq(r)}$ is small when $q(r)$ is small and decays rapidly when $q(r)$ is large. For any threshold $\tau \in (0,1)$, we split the sum into contributions from "large" and "small" probabilities:

$$\sum_r q(r)e^{-2Kq(r)} \leq e^{-2K\tau} + \sum_{r:q(r)\leq\tau} q(r).$$

Thus, to ensure $\mathbb{E}_q[(1-q(r))^{2K}] \leq \varepsilon$, it suffices to choose τ such that $\sum_{r:q(r)\leq\tau} q(r) \leq \varepsilon/2$ and set

$$K \geq \frac{1}{2\tau}\log\left(\frac{2}{\varepsilon}\right).$$

Robustness to oracle and sampling perturbations. The oracle step (Step 6) in Algorithm 1 may appear restrictive. In practice, it can be relaxed by replacing the exact oracle with a best-arm identification (BAI) procedure; related works are discussed in Supplementary A. When $\mathcal{A}_{\text{full}}$ is finite, classical BAI methods (e.g., (Audibert & Bubeck, 2010)) can identify the optimal arm with arbitrarily high probability using a finite number of samples.

We also allow for mild misspecification in the sampling distribution. We show that both perturbations introduce only controlled additive errors in the regret bound.

Corollary 4.7 (Approximate oracle via BAI)**.** *Under the same conditions as Theorem 4.4, suppose $\mathcal{A}_{\text{full}}$ is large but finite, and the oracle step in Algorithm 1 is replaced by a BAI procedure that identifies the optimal arm $a^*(\theta)$ with probability at least $1-\delta$. Then*

$$\mathbb{E}_{\theta,\mathcal{A}}[\text{Regret}] \leq \max_{\ell\leq m} \mathbb{E}_\theta\left[\max_{a\in r_\ell}(\mu_a - \mu_{a_\ell})\right] + C\epsilon\sqrt{\log m} + \left(\mathbb{E}_q\left[(1-(1-\delta)\,q(r))^{2K}\right]\cdot \mathbb{E}_\theta\left[\max_{a\in\mathcal{A}_{\text{full}}} \mu_a^2\right]\right)^{1/2},$$

where $C > 0$ is an absolute constant.

Corollary 4.8 (Misspecified sampling distribution)**.** *Under the same conditions as Theorem 4.4, suppose that during the K sampling steps in Algorithm 1, the sampling distribution is misspecified as $\tilde{q}(r_\ell) \geq q(r_\ell) - \kappa$ for all $\ell \leq m$, with $\kappa > 0$. Then*

$$\mathbb{E}_{\theta,\mathcal{A}}[\text{Regret}] \leq \max_{\ell\leq m} \mathbb{E}_\theta\left[\max_{a\in r_\ell}(\mu_a - \mu_{a_\ell})\right] + C\epsilon\sqrt{\log m} + \left(\mathbb{E}_q\left[(1-q(r)+\kappa)^{2K}\right]\cdot \mathbb{E}_\theta\left[\max_{a\in\mathcal{A}_{\text{full}}} \mu_a^2\right]\right)^{1/2},$$

where $C > 0$ is an absolute constant.

Proof. The proof closely follows that of Theorem 4.4. The only modification is in the upper bound on the probability of missing a cluster r_ℓ after K samples. Let $\mathcal{A}$ denote the algorithm output.

For Corollary 4.7, we have

$$\Pr[\mathcal{A} \cap r_\ell = \emptyset] \;\le\; (1 - (1-\delta)\, q(r_\ell))^K .$$

For Corollary 4.8, we have

$$\Pr[\mathcal{A} \cap r_\ell = \emptyset] \;\le\; (1 - (q(r_\ell) - \kappa))^K .$$

Substituting these bounds into the proof of Theorem 4.4 yields the desired results. □

Remark 4.9 (Linear Bandits and Known Gaussian Process Regret Bounds). As discussed in Section 3.1, when dimension n is fixed, we can view the problem setup (5) as that of Bayesian linear bandits. In Section 1.1 we have mentioned two motivations for the action subset selections problem: The intrinsic interest in reducing the size of the action set for purposes such inventory maintenance, and a possibility of faster learning in a new bandit instances, when the action set is reduced from $\mathcal{A}_{\text{full}}$ to $\mathcal{A}$, see (2). While the first scenario holds for linear bandits as well, the second scenario may be weaker, since in linear bandits the regret may be bounded either as $O(\sqrt{n \cdot \log|\mathcal{A}_{\text{full}}|}\sqrt{T})$, (Russo & Van Roy, 2016), or as $O(n\sqrt{T})$, (Dani et al., 2008). Thus, in these situations the regret depends at most only logarithmically on the size of $|\mathcal{A}_{\text{full}}|$, *if* the dimension n is known to be small. However, as discussed earlier, we do not require the dimension to be small, and our bounds depend on the actual geometry of the set instead of relying on global constraints such as the dimension. Thus our bounds may remain small even when n is arbitrarily large.

Similar considerations also apply more generally to existing regret bounds for Bayesian bandits with Gaussian Process reward distributions (a problem also known as Bayesian Optimization), (Srinivas et al., 2009; Garnett, 2023). While these bounds are applicable with infinite $\mathcal{A}_{\text{full}}$, they involve global parameters of the model and assumptions on the kernel. For instance, for linear kernels the bound is $O(n\sqrt{T})$, while for RBF kernels one has $O(\sqrt{T\,(\log T)^{n+1}})$ - exponential in the underlying dimension, (Srinivas et al., 2009). Thus such bounds do not reflect the underlying clustering or non-linear low dimensionality of the data.

Even for small n, each round incurs non-trivial optimization costs. Both UCB and TS require solving an inner maximization over $\mathcal{A}_{\text{full}}$, whose difficulty depends on the action space structure. In linear bandits, UCB solves $\max_{a\in\mathcal{A}_{\text{full}}} \max_{\theta\in\mathcal{C}_t} \langle\theta, a\rangle$, which is NP-hard in general—even for convex $\mathcal{A}_{\text{full}}$ (Zhang et al., 2025)—but trivial for small finite sets. TS similarly requires $\max_{a\in\mathcal{A}_{\text{full}}} \langle\tilde{\theta}_t, a\rangle$; while linear in a, this is costly for large finite sets, though near-sublinear costs $\tilde{O}(|\mathcal{A}_{\text{full}}|^{1-o(1)})$ are achievable under structural assumptions (Jun et al., 2017; Yang et al., 2022). For GP bandits, the inner problem maximizes a non-concave acquisition function (e.g., GP-UCB), which is generally intractable for continuous domains. With finite $\mathcal{A}$ of size $|\mathcal{A}|$, the per-step cost is $\tilde{O}(|\mathcal{A}|h^2 + h^3)$ for batch size h (Calandriello et al., 2022). Continuous domains typically require discretizations for approximate maximization (Chowdhury & Gopalan, 2017; Salgia et al., 2021).

Finally, as discussed earlier, while Theorem 4.4 is stated for a fixed partition, for any $\epsilon > 0$ we can construct a partition derived from an ϵ-net, and to obtain a corresponding regret bound. We carry this construction out in the following result. Recall the covering number $N(\mathcal{A}_{\text{full}}, \epsilon)$ from (7).

Theorem 4.10. *Set $M := \operatorname{diam}(\mathcal{A}_{\text{full}})$ and fix $\epsilon > 0$. Let $\mathcal{A}$ be the output of Algorithm 1. Then, for*

$$K \ge c \cdot (M/\epsilon)^2 \cdot N(\mathcal{A}_{\text{full}}, \epsilon), \tag{10}$$

we have

$$\mathbb{E}_{\theta,\mathcal{A}}[\text{Regret}] \le C\epsilon \min\{\sqrt{n}, \sqrt{\log|\mathcal{A}_{\text{full}}|}\} + C'\epsilon\sqrt{\log N(\mathcal{A}_{\text{full}}, \epsilon)},$$

where $C, C', c > 0$ are absolute constants.

In particular, by taking $\epsilon \to 0^+$, and taking K to satisfy (10), we have $\mathbb{E}_{\theta,\mathcal{A}}[\text{Regret}] \to 0$.

We now present a general lower bound that holds for any algorithm.

Theorem 4.11. *For any algorithm that outputs a subset $\mathcal{A}$ of cardinality K, the expected regret satisfies*

$$\mathbb{E}_\theta[\text{Regret}] \;\ge\; \max_{\ell\le m} \mathbb{E}_\theta\left[\max_{a\in r_\ell} \mu_a - \mu_{a_\ell}\right] - C\;\Delta(\mathcal{A})\sqrt{\log m + \log K},$$

for some absolute constant $C > 0$.

Proof sketch. The proof closely mimics that of Theorem 4.3. Define $Y_{\ell,i} := \sup_{a\in r_\ell\cup\{a_i\}}(\mu_a - \mu_{a_i})$. When $a^*(\theta) \in r_\ell$, $\text{Regret} = \min_i \max_\ell Y_{\ell,i}$. Decompose $Y_{\ell,i}$ into mean and fluctuation; the latter is uniformly controlled by Gaussian concentration, giving a $\Delta(\mathcal{A})\sqrt{\log mK}$ term. Finally, a min–max argument lower bounds $\min_i \max_\ell \mathbb{E}Y_{\ell,i}$ by $\max_\ell \mathbb{E}[\max_{a\in r_\ell}(\mu_a - \mu_{a_\ell})]$. □

Remark 4.12. The leading term in Theorem 4.11,

$$\max_{\ell\le m} \mathbb{E}_\theta\left[\max_{a\in r_\ell}(\mu_a - \mu_{a_\ell})\right],$$

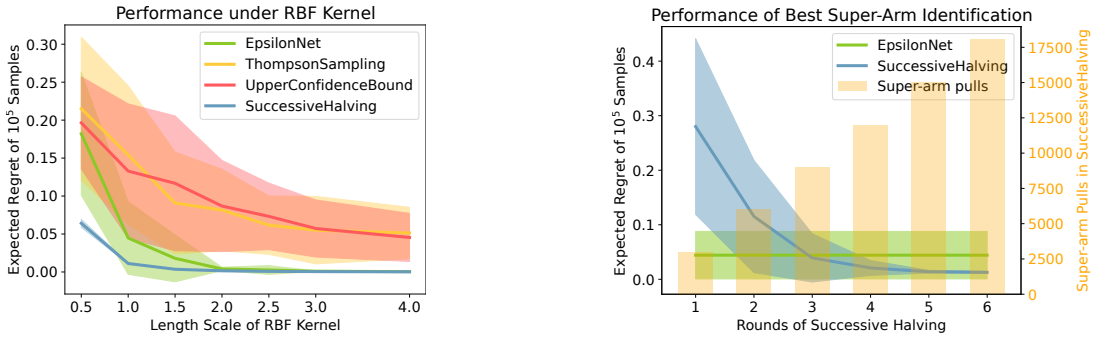

Figure 2. Comparison on solving (11) by selecting $K = 5$ actions from 15 grid points in $[0, 2]$, with outcome functions $f(a) = \mu_a$ sampled via an RBF kernel at varying length-scales. **Left:** Expected regret over 50 repetitions: our method with exhaustive search (green) vs. super-arm TS (yellow), UCB (orange), and Successive Halving (SH; blue). TS/UCB run for 3,000 rounds; SH uses a budget of 37,000 pulls. **Right:** SH's expected regret (blue) and cumulative super-arm pulls (yellow bars) per round vs. our method (green). SH requires nearly 10,000 pulls—about three times the number of super-arms—to match our performance.

coincides with the cluster-complexity term appearing in the upper bound (Theorem 4.4) and captures the intrinsic difficulty of the problem. The remaining term

$$C\Delta(\mathcal{A})\sqrt{\log m + \log K}$$

appears as a subtractive slack term in the lower bound. In particular, the dependence on K is only logarithmic Moreover, when the output $\mathcal{A}$ is restricted to be a reference subset, so that $K = m$, this lower bound recovers the same qualitative scaling as the reference-subset lower bound in Theorem 4.3, up to constants.

4.3. Regret Bound for Sub-Gaussian Process

The Gaussian process assumption can be relaxed to a sub-Gaussian one, such that the assumption in Equation (5) is lifted. Instead, we only require that $\mathbb{E}_\theta[\mu_a(\theta)] = 0$ for all $a \in \mathcal{A}_{\text{full}}$, and that the random process $\mu_a a \in \mathcal{S}$ satisfies the following increment condition:

$$\forall u > 0, \quad \Pr\left[|\mu_a - \mu_{a'}| \geq u\right] \leq 2\exp\left(-\frac{u^2}{2\|a - a'\|_2^2}\right).$$

Let $\gamma_2(\mathcal{S})$ denote Talagrand's chaining functional (Talagrand, 2014) for a set $\mathcal{S} \subset \mathbb{R}^n$, equipped with the Euclidean norm. It is the sharpest known bound that for a sub-Gaussian process indexed by $\mathcal{S}$.

Using this functional, we establish a regret bound for our algorithm:

Theorem 4.13. *Let $\{\mu_a\}_{a\in\mathcal{A}_{\text{full}}}$ be a mean-zero sub-Gaussian process. Let $\mathcal{A}$ be output of Algorithm 1. Consider a partition $\mathcal{R} := \{r_\ell\}_{\ell\leq m}$ of the full action space, where $m \geq 2$. Then,*

$$\mathbb{E}_{\theta,\mathcal{A}}[\text{Regret}] \leq 8 \max_{\ell\leq m} \gamma_2(r_\ell) + 57\,\epsilon\sqrt{\log(m)} + \left(\mathbb{E}_q\left[(1-q(r))^{2K}\right] \cdot \mathbb{E}_\theta\left[\max_{a\in\mathcal{A}_{\text{full}}} \mu_a^2\right]\right)^{1/2}.$$

5. Generalization and Empirical Validation

We now treat the expected rewards $\{\mu_a\}_{a\in\mathcal{A}_{\text{full}}}$ as evaluations of an unknown outcome function $f : \mathcal{A}_{\text{full}} \to \mathbb{R}$ over the abstract action set, where $f(a) := \mu_a$. Although our framework is general, we instantiate it in the experiments using outcome functions from an RKHS (Williams & Rasmussen, 2006), a function class for which our method is applicable and theoretically sound. The **synthetic outcome functions** are sampled from a Gaussian process with an RBF kernel (with length-scale parameter ℓ), yielding outcome functions consistent with an underlying RKHS (Kanagawa et al., 2018). The **action space** corresponds to the input domain of the RBF kernel, while the kernel itself is not revealed to the learner.

Details justifying the applicability of our framework to RKHS, as well as the data generation procedure, are provided in Supplementary B.

Returning to the objective stated in the Introduction, we seek a subset $\mathcal{A} \subseteq \mathcal{A}_{\text{full}}$ of cardinality K that minimizes the expected regret defined in (1). Since $\mathbb{E}[\max_{a\in\mathcal{A}_{\text{full}}} \mu_a]$ is constant with respect to $\mathcal{A}$, this is equivalent to solving:

$$\max_{\mathcal{A}\subseteq\mathcal{A}_{\text{full}}} \mathbb{E}\left[\max_{a\in\mathcal{A}} \mu_a\right] \quad \text{subject to } |\mathcal{A}| = K. \tag{11}$$

This problem is generally non-trivial: the objective involves an expectation over random variables with an unspecified (and potentially correlated) distribution, and the inner max operator induces non-smooth dependence on $\mathcal{A}$. Nevertheless, our method efficiently circumvents the combinatorial complexity inherent in (11).

5.1. Comparisons at the Super-Arm Level

We compare our method against TS and UCB in terms of super-arm identification performance: we aim to identify near-optimal subsets of (11) by selecting $K = 5$ ac-

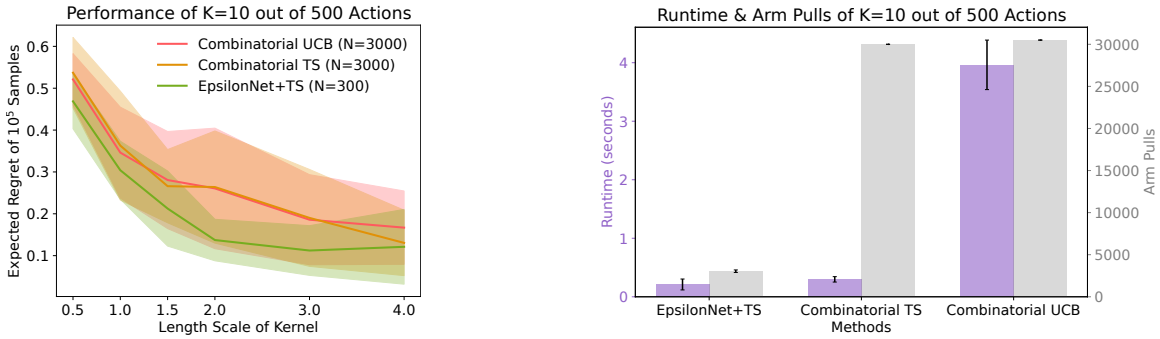

Figure 3. Comparison of EpsilonNet+TS (green), CTS (brown), and CUCB (red) on solving (11) with $K = 10$ actions from 500 grid points in $[-5, 5]$, using outcome functions $f(a) = \mu_a$ sampled via an RBF kernel at varying length-scales. In EpsilonNet+TS, the `argmax` oracle in Algorithm 1 is replaced by TS; CTS and CUCB run for 3,000 rounds. All methods use a $\mathcal{N}(0, 1)$ prior. **Left:** expected regret (30 repetitions). **Right:** runtime (purple, left axis) and arm pulls (gray, right axis).

tions from an action space consisting of 15 grid points in $[0, 2]$. Each 5-action tuple is treated as a super-arm. This yields $\binom{15}{5} = 3003$ super-arms—a moderate number that allows us to approximate the optimal super-arm using Successive Halving (SH), although SH becomes computationally infeasible for larger action spaces. Outcome functions are sampled from via an RBF kernel at length scales $l \in \{0.5, 1, \ldots, 4\}$.

In our approach, we run Algorithm 1 until $K = 5$ distinct actions are selected, using exhaustive search over the action space to compute the optimal action $a^*(\theta)$ for each bandit instance θ. TS and UCB operate in a bandit feedback setting: at each round (i.e., for a new θ), a super-arm $\mathcal{A}$ is selected, the payoff $\max_{a \in \mathcal{A}} \mu_a(\theta)$ is observed, and the policy is updated. We run both methods for 3,000 rounds—matching the total number of super-arms ($\binom{15}{5} = 3003$). The payoff of a super-arm $\mathcal{A}$ is defined as $\max_{a \in \mathcal{A}} \mu_a(\theta)$ to align with the objective in (11); thus, these methods directly optimize $\mathbb{E}[\max_{a \in \mathcal{A}} \mu_a]$. Since this payoff is unbounded, we use a $\mathcal{N}(0, 1)$ prior for both TS and UCB. For SH, we allocate a fixed budget of $N \log_2 N \approx 37{,}000$ arm pulls (following (Karnin et al., 2013)), where $N = 3003$ is the number of super-arms. In each round (corresponding to a new θ), the set of surviving super-arms is halved based on empirical payoffs, continuing until one remains or the budget is exhausted. Finally, we estimate the expected regret of all methods over 10^5 additional sampled outcome functions.

The left subplot of Figure 2 shows the expected regret of our method (green), TS (yellow), UCB (orange), and SH (blue). Solid curves and shaded regions denote the mean $\pm$ one standard deviation over 50 repetitions. As observed, the expected regret decreases with increasing length-scale l. The length-scale l controls the effective number of independent actions: for $|a - a'| \gg l$, μ_a and $\mu_{a'}$ become nearly uncorrelated. In the limit of large l, the outcomes $\{\mu_a\}_{a \in \mathcal{A}_{\text{full}}}$ behave like i.i.d. $\mathcal{N}(0, 1)$ variables.

The right subplot of Figure 2 highlights SH's impracticality. We ran both our method (green) and SH (blue) for 50 repetitions with fixed $l = 1$, tracking the best super-arm found and total super-arm pulls after each SH round. The blue curves (with error shades) show SH's expected regret (mean $\pm$ std.), while the yellow bars (right y-axis) indicate cumulative super-arm pulls per round. SH matches our performance only after three rounds—but at roughly three times the total number of super-arms.

5.2. Oracle-Free Comparisons with CTS and CUCB

While Algorithm 1 relies on an oracle `argmax` step, this is often infeasible in practice. We replace it with a simple TS rule, yielding a fully implementable, oracle-free variant (EpsilonNet+TS). We compare this method against combinatorial TS (CTS) and combinatorial UCB (CUCB)—two practical approaches for choosing an action subset with $|\mathcal{A}| = K$—under the same setup as in Section 5.1, but now selecting $K = 10$ actions from an action space consisting of 500 grid points over $[-5, 5]$.

Our approach runs Algorithm 1 until 10 distinct actions are identified. The oracle `argmax` step is approximated by running TS for 300 rounds; the action chosen in the final round serves as an approximation of $a^*(\theta)$. CTS and CUCB operate in a semi-bandit setting for 3000 rounds. In each round t, an instance θ_t is sampled, an action subset $\mathcal{A}_t$ of cardinality $K = 10$ is selected, and rewards $\mu_a(\theta_t)$ are observed for all $a \in \mathcal{A}_t$. The subset selected in the final round is taken as the algorithm's output. All methods use a $\mathcal{N}(0, 1)$ prior, consistent with the ground truth.

The left subplot of Figure 3, the left subplot shows the expected regret (mean $\pm$ standard deviation over 30 repetitions) for EpsilonNet+TS (green), CTS (brown), and CUCB (red). As observed, the poor performance of CTS and CUCB aligns with discussions in related work. The right subplot reports runtime and arm pulls across 30 repetitions with fixed

$l = 1$: the left axis (purple) shows average runtime $\pm$ one standard deviation, while the right axis (gray) shows total base arm pulls. EpsilonNet+TS uses exactly 3,000 arm pulls (300 TS rounds each $\times$ 10 base arms). CTS incurs comparable runtime but requires far more pulls (3,000 CTS rounds $\times$ 10 base arm pulls). CUCB is significantly slower and incurs additional pulls due to its mandatory initialization.

6. Conclusions and Future Work

We proposed a framework for selecting a small representative subset of actions from a very large shared action space in a family of bandit problems. Our simple subset-selection algorithm exploits reward correlations to identify representative actions with controllable performance loss. Natural directions for future work include developing data-dependent stopping rules for choosing the subset size K, and extending the setting from bandits to Markov decision processes to capture sequential decision making.

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Supplementary Summary: The appendix is divided into four parts.

First, Sections A-B provide additional related work, and experimental details omitted from the main text due to space constraints.

Second, Sections C to I contain the proofs of all the theorems and lemmas stated both in the main body.

Third, Sections J-K present auxiliary tools used throughout the paper.

Fourth, Sections L-N include additional examples and a numerical study of Algorithm 1. This part features a low-dimensional ($n = 3$) action space on the unit sphere to investigate the effect of cluster diameter on expected regret, in line with Theorem 4.2. It also studies two extensions of the varying dependence structure of actions introduced in the main body:

- Nonstationary dependence over the action space: Using a nonstationary Gibbs kernel, where the kernel value depends not only on the difference between two inputs a, a' but also on the location of a.
- Limiting case: The collection of expected outcomes $\{\mu_a\}_{a\in\mathcal{A}_{\text{full}}}$ becomes i.i.d.

LLM Usage: Large language models are used for polishing writing, and finding relevant research.

A. Extra Related Works

Multi-Armed Bandits (Lai & Robbins, 1985; Auer et al., 2002a) is defined by a set of actions (arms), each deliver outcomes that are independently drawn from a fixed and unknown distribution. The decision-maker sequentially selects an action, observes its outcome, and aims to maximize cumulative outcomes over time. Popular methods include the UCB (Auer et al., 2002a), TS (Agrawal & Goyal, 2012), and EXP3 (Auer et al., 2002b) for adversarial settings.

Optimal Action Identification focuses on identifying the action with the highest expected outcome in a MAB setting using as few samples as possible (Jamieson & Nowak, 2014; Kaufmann et al., 2016). Popular methods in fixed confidence setting include Action Elimination (Even-Dar et al., 2006; Karnin et al., 2013), UCB, and LUCB, all of which achieve sample complexity within a $\log(|\mathcal{A}_{\text{full}}|)$ factor of the optimum. In fixed budget setting, there is Successive halving (Karnin et al., 2013), successive reject (Audibert & Bubeck, 2010).

Linear Bandits assumes that the expected outcome of each action depends *linearly* on the inner product between an unknown parameter vector θ and the feature representation of an action $a \in \mathcal{A}_{\text{full}}$ (Auer, 2002; Dani et al., 2008; Rusmevichientong & Tsitsiklis, 2010). This line of work assumes that action feature vectors are known *a priori*, so the cardinality of the action space is not a limiting factor and may even be infinite. The bound scales as $O(n\sqrt{T})$, where n is the dimension of the feature space.

GP Bandits address the case where explicit feature representations are unavailable, but outcomes are assumed to be correlated according to a known kernel function $k(\cdot,\cdot)$ (Williams & Rasmussen, 2006). Srinivas et al. (2009) model the reward function as a sample from a Gaussian process prior and propose the GP-UCB algorithm, which achieves cumulative regret $O\big(\sqrt{T\,\gamma_T \log|\mathcal{A}_{\text{full}}|}\big)$ for finite action spaces, where γ_T denotes the maximum information gain after T rounds. The widespread adoption of this approach—both for finite action sets (Valko et al., 2013; Li & Scarlett, 2022) and continuous domains (Chowdhury & Gopalan, 2017) highlights the practicality of assuming that action outcomes are correlated.

Epsilon Nets have two standard definitions. The first, geometric definition (Vershynin, 2018), requires that radius-ϵ balls centered at net points cover the set. It relates to the covering number and extends to function classes, as in Russo & Van Roy (2013). The second, measure-theoretic definition (Matousek, 2013), requires the net to intersect all subsets of sufficiently large measure. The classic ϵ-net algorithm by Haussler & Welzl (1986) remains the simplest and most broadly applicable method. Later works aim to reduce net size (Pach & Tardos, 2011; Rabani & Shpilka, 2009; Mustafa, 2019) and address online settings (Bhore et al., 2024).

Expected supremum of Gaussian process for a given set $\mathcal{S}$ refers to the term $\mathbb{E}\left[\max_{a\in\mathcal{S}} \mu_a\right]$. It is an important topic in high-dimensional probability (Vershynin, 2018) and asymptotic convex geometry (Rothvoss, 2021, Chapter 9). The sharpest known bounds are due to (Talagrand, 2014). The expectation of regret defined in (1) is indeed the difference between the expected suprema of two Gaussian processes, $\{\mu_a\}_{a\in\mathcal{A}\text{full}}$ and $\{\mu_a\}_{a\in\mathcal{A}}$. This connection motivates our ideas and analysis around the concept of an ϵ-net, which is a central tool for bounding the expected supremum.

B. Extra Details of Experiments

In general cases, the decision-maker may not have explicit access to the full structure of the action space, especially in high-dimensional settings. Instead, they are given a list of actions and can observe the expected outcomes of these actions through sampling. We therefore treat the expected outcomes $\{\mu_a\}_{a\in\mathcal{A}_{\text{full}}}$ as a family of random variables indexed by an abstract set. In a single bandit, these expected outcomes define an **outcome function** f over the action space $\mathcal{A}_{\text{full}}$: $f(a) := \mu_a, \forall a \in \mathcal{A}_{\text{full}}$. We show that our framework applies when the outcome functions lie in a reproducing kernel Hilbert space (RKHS) (Williams & Rasmussen, 2006). Consider a positive semidefinite kernel $k : \mathcal{A}_{\text{full}} \times \mathcal{A}_{\text{full}} \to \mathbb{R}$. Consider a positive semidefinite kernel $k : \mathcal{A}_{\text{full}} \times \mathcal{A}_{\text{full}} \to \mathbb{R}$. By Mercer's theorem, for a non-negative measure $\mathbb{P}$ over $\mathcal{A}_{\text{full}}$, if the kernel satisfies $\int_{\mathcal{A}_{\text{full}}\times\mathcal{A}_{\text{full}}} k^2(a,a')d\mathbb{P}(a)d\mathbb{P}(a') < \infty$, then it admits an eigenfunction expansion: $k(a,a') = \sum_{i\le\infty} \lambda_i\phi_i(a)\phi_i(a')$, where $(\phi_i)_{i\le\infty}$ are orthonormal eigenfunctions under $\mathbb{P}$, and $(\lambda_i)_{i\le\infty}$ are non-negative eigenvalues. Let outcome function$f$ be of the form $f(\cdot) = \sum_{i=1}^{N} \alpha_i k(\cdot, a_i)$ for some integer $N \ge 1$, and a set of points $\{a_i\}_{i=1}^{N} \subset \mathcal{A}_{\text{full}}$ and a weight vector $\alpha \in \mathbb{R}^N$. The function f can be rewritten as

$$\mu_a = f(a) = \langle \mathbf{f}, \Phi(a)\rangle,$$

where $\mathbf{f}$ is a vector of coefficients, and $\Phi(a)$ usually refereed as a feature map, has entries $\Phi_i(a) = \sqrt{\lambda_i}\phi_i(a)$. The full action space is the one formed by the feature vectors $\Phi(a)$ for $a \in \mathcal{A}_{\text{full}}$.

Data Generation: Assuming the outcome function is sampled from a kernel implies action correlations. To study varying correlation levels, we use the RBF kernel

$$k(a,a') := \exp\left(-\frac{\|a-a'\|^2}{2l^2}\right),$$

where the length-scale l controls dependence among actions. Following (Kanagawa et al., 2018) (Theorem 4.12), we sample functions from the RKHS by forming the kernel matrix $\mathbf{K}_{a,a'} = k(a,a')$ over $\mathcal{A}_{\text{full}}$ and drawing $f \sim \mathcal{N}(0,\mathbf{K})$. Although the kernel generates the outcomes, it is not revealed to the learner; thus, actions are treated as abstract indices rather than kernel inputs.

C. Proof of Lemma 3.3

Statement: Given a partition $\mathcal{R}$ of the full action space, and the importance measure q assigning a value to each cluster $r \in \mathcal{R}$. Let $\mathcal{A}$ be the output of Algorithm 1 after K samples. Then, with probability at least $1 - \frac{1}{\epsilon}\exp(-K\epsilon)$, it holds that for any cluster $r \in \mathcal{R}$,

$$r \cap \mathcal{A} \neq \emptyset \quad \text{whenever} \quad q(r) > \epsilon.$$

Proof. By definition, the algorithm draws K i.i.d. samples from clusters in $\mathcal{R}$ according to the distribution q. Define a set of typical clusters $R_\epsilon := \{r \in \mathcal{R} : q(r) > \epsilon\}$. Then,

$$\Pr[r \cap \mathcal{A} = \emptyset] = (1-q(r))^K < (1-\epsilon)^K \le \exp(-K\epsilon),\ \forall r \in R_\epsilon, \tag{12}$$

where the equality is the probability of missing the cluster r for K times, the first inequality uses the definition of R_ϵ. The last inequality uses $\log(1-\epsilon)^K = K\log(1-\epsilon)$ and the inequality $\log(1-\epsilon) \le -\epsilon$ for $\epsilon < 1$ (TOPSØE[1], 2007). Therefore,

$$\begin{aligned}
&\Pr[r \cap \mathcal{A} \neq \emptyset \text{ whenever } q(r) > \epsilon] \\
=&1 - \Pr[r \cap \mathcal{A} \neq \emptyset\ \exists r \in R_\epsilon] \\
\ge&1 - \sum_{r\in R_\epsilon} \Pr[r \cap \mathcal{A} \neq \emptyset] \\
>&1 - \sum_{r\in R_\epsilon} \exp(-K\epsilon) \\
=&1 - |R_\epsilon|\exp(-K\epsilon) \ge 1 - 1/\epsilon \cdot \exp(-K\epsilon),
\end{aligned}$$

where the first inequality uses union bound, the second inequality uses (12). The last inequality uses the fact that there could be at most $\lfloor 1/\epsilon \rfloor$ clusters in R_ϵ. □

D. Proof of Theorem 4.2

To bound the expected regret of reference subsets, we need a few more lemmas:

Lemma D.1 (Expectation integral identity)**.** *Given a non-negative random variables X. If $\Pr[X \geq u] \leq c \exp\left(-\frac{u^2}{\epsilon^2}\right)$ for any $u > 0$, then $\mathbb{E}X \leq C\epsilon\sqrt{\log c}$, where ϵ, c, C are positive constants.*

Proof.

$$
\begin{aligned}
\mathbb{E}X &= \int_0^\infty \Pr[X \geq u]\, du \\
&= \int_0^{u_0} \Pr[X \geq u]\, du + \int_{u_0}^\infty \Pr[X \geq u]\, du \\
&\leq u_0 + \frac{1}{u_0} \int_{u_0}^\infty u \cdot \Pr[X \geq u]\, du \\
&\leq u_0 + \frac{c}{u_0} \int_{u_0}^\infty u \cdot \exp\left(-\frac{u^2}{\epsilon^2}\right) du \\
&= u_0 + \exp\left(-\frac{u_0^2}{\epsilon^2}\right) \cdot \frac{c\epsilon^2}{2u_0} \\
&= \epsilon\sqrt{\log c} + \frac{\epsilon}{2\sqrt{\log c}} \leq C\epsilon\sqrt{\log c},
\end{aligned}
$$

where the first equality uses integrated tail formula of expectation (cf. Lemma 1.6.1 of (Vershynin, 2018)). The last equality set $u_0 := \epsilon\sqrt{\log c}$. □

Lemma D.2 (Borell-TIS inequality; Lemma 2.4.7 of (Talagrand, 2014))**.** *Given a set $\mathcal{S}$, and a zero-mean Gaussian process $(X_a)_{a\in\mathcal{S}}$. Let $\epsilon := \sup_{a\in\mathcal{S}} \left(\mathbb{E}X_a^2\right)^{\frac{1}{2}}$. Then for $u > 0$, we have*

$$
\Pr\left[\left|\sup_{a\in\mathcal{S}} X_a - \mathbb{E}\sup_{a\in\mathcal{S}} X_a\right| \geq u\right] \leq 2\exp\left(-\frac{u^2}{2\epsilon^2}\right).
$$

It means that the size of the fluctuations of $\sup_{a\in\mathcal{S}} X_a$ is governed by the size of the individual random variables X_a.

Statement of Theorem 4.2: Consider a partition $\mathcal{R} := \{r_\ell\}_{\ell\leq m}$ of the full action space, with $\epsilon := \max_{r\in\mathcal{R}} \operatorname{diam}(r)$, and an arbitrary reference subset $\mathcal{A}$. Then, for some constant $C > 0$,

$$
\mathbb{E}_\theta[\text{Regret}] \leq \max_{r\in\mathcal{R}} \mathbb{E}_\theta\left[\max_{a\in r} \mu_a - \mu_{a_\ell}\right] + C\epsilon\sqrt{\log m} \;=\; \max_{r\in\mathcal{R}} \mathbb{E}_\theta\left[\max_{a\in r} \mu_a\right] + C\epsilon\sqrt{\log m}.
$$

Proof. Fix ℓ, define a Gaussian process $\{Z_a\}_{a\in r_\ell}$ where $Z_a := \mu_a - \mu_{a_\ell}$. Define a non-negative random variable

$$
Y_\ell := \sup_{a\in r_\ell} Z_a = \sup_{a\in r_\ell} \mu_a - \mu_{a_\ell}.
$$

Since $\mathbb{E}\mu_a = 0$ for all $a \in \mathcal{A}_{\text{full}}$, it holds that $\mathbb{E}Y_\ell = \mathbb{E}\max_{a\in r_\ell} \mu_a$.

Further, by definition $r_\ell \subset B(a_\ell, \epsilon)$, such that $\mathbb{E}Z_a^2 = \mathbb{E}(\mu_a - \mu_{a_\ell})^2 = \|a - a_\ell\|_2^2 \leq \epsilon^2$. Using Lemma D.2 on the process $\{Z_a\}_{a\in r_\ell}$, we have

$$
\Pr\left[|Y_\ell - \mathbb{E}Y_\ell| \geq u\right] \leq 2\exp\left(-\frac{u^2}{2\epsilon^2}\right).
$$

By union bound, we have

$$
\Pr\left[\max_{\ell\leq m} |Y_\ell - \mathbb{E}Y_\ell| \geq u\right] \leq 2m\exp\left(-\frac{u^2}{2\epsilon^2}\right).
$$

Using Lemma D.1, we have for some absolute constant $C > 0$

$$\mathbb{E} \max_{\ell \le m} |Y_\ell - \mathbb{E}Y_\ell| \le C\epsilon\sqrt{\log m}. \tag{13}$$

Upper bound:

When the cluster r_ℓ contains the optimal action, i.e., $a^*(\theta) \in r_\ell$, we have

$$\text{Regret} = \max_{a \in \mathcal{A}_{\text{full}}} \mu_a - \max_{a' \in \mathcal{A}} \mu_{a'} = \max_{a \in r_\ell} \mu_a - \max_{a' \in \mathcal{A}} \mu_{a'} \le \max_{a \in r_\ell} \mu_a - \mu_{a_\ell} = Y_\ell \le \max_{\ell' \le m} Y_{\ell'}, \tag{14}$$

where the first equality follows from the definition of regret in (1), and the second equality follows from the assumption that $a^*(\theta) \in r_\ell$. The inequality holds because $a_\ell \in \mathcal{A}$, and hence $\max_{a \in \mathcal{A}} \mu_a \ge \mu_{a_\ell}$. The final equality follows from the definition of Y_ℓ. Therefore,

$$\mathbb{E}\,\text{Regret} \le \mathbb{E} \max_{\ell \le m} Y_\ell \le \max_{\ell \le m} \mathbb{E}Y_\ell + \mathbb{E} \max_{\ell \le m} |Y_\ell - \mathbb{E}Y_\ell| \le \max_{\ell \le m} \mathbb{E} \max_{a \in r_\ell} \mu_a + C\epsilon\sqrt{\log m}, \tag{15}$$

where the first inequality uses (14). The second inequality uses $\max_{\ell \le m} Y_\ell \le \max_{\ell \le m} \mathbb{E}Y_\ell + \max_{\ell \le m} |Y_\ell - \mathbb{E}Y_\ell|$, since $Y_\ell \le \mathbb{E}Y_\ell + |Y_\ell - \mathbb{E}Y_\ell|$. The last inequality uses (13) and the identity $\mathbb{E}[Y_\ell] = \mathbb{E}[\max_{a \in r_\ell} \mu_a]$. □

E. Proof of Theorem 4.3

The proof closely follows that of the upper bound, but introduces auxiliary variables $Y_{\ell,\ell'}$ indexed by pairs of clusters (allowing $\ell = \ell'$), instead of the single-index variables Y_ℓ used in the upper bound. This modification is necessary because, when $a^*(\theta) \in r_\ell$, the argument must compare against the best representative $a_{\ell'}$, i.e., the one attaining $\max_{\ell' \le m} \mu_{a_{\ell'}}$.

We define the outer radius induced by an action subset $\mathcal{A}$ as

$$\Delta(\mathcal{A}) := \max_{a \in \mathcal{A}} \sup_{a' \in \mathcal{A}_{\text{full}}} \|a - a'\|_2,$$

which satisfies $\Delta(\mathcal{A}) \le \text{diam}(\mathcal{A}_{\text{full}})$.

Statement of Theorem 4.3: Consider a partition $\mathcal{R} := \{r_\ell\}_{\ell \le m}$ of the full action space, with an arbitrary reference subset $\mathcal{A}^\epsilon$. Then, for some absolute constant $C > 0$,

$$\mathbb{E}_\theta[\text{Regret}] \;\ge\; \max_{\ell \le m} \mathbb{E}\left[\max_{a \in r_\ell} \mu_a - \mu_{a_\ell}\right] - C\,\Delta(\mathcal{A}^\epsilon)\sqrt{\log m}.$$

Proof. **Step 1:** Fix ℓ, ℓ' (allowing $\ell = \ell'$), define a centered Gaussian process $\{Z_a\}_{a \in r_\ell \cup \{a_{\ell'}\}}$ where $Z_a := \mu_a - \mu_{a_\ell}$. Define a non-negative random variable

$$Y_{\ell,\ell'} := \sup_{a \in r_\ell \cup \{a_{\ell'}\}} Z_a = \sup_{a \in r_\ell \cup \{a_{\ell'}\}} \mu_a - \mu_{a_{\ell'}}.$$

Step 2: Set $\Delta := \Delta(\mathcal{A}^\epsilon)$. Further, by definition of Δ, we have $\mathbb{E}Z_a^2 = \mathbb{E}(\mu_a - \mu_{a_\ell})^2 = \|a - a_\ell\|_2^2 \le \Delta^2$. Using Lemma D.2 on the process $\{Z_a\}_{a \in r_\ell \cup \{a_{\ell'}\}}$, we have

$$\Pr\left[|Y_{\ell,i} - \mathbb{E}Y_{\ell,\ell'}| \ge u\right] \le 2\exp\left(-\frac{u^2}{2\Delta^2}\right).$$

By union bound, we have

$$\Pr\left[\max_{\ell,\ell' \le m} |Y_{\ell,i} - \mathbb{E}Y_{\ell,\ell'}| \ge u\right] \le 2m^2 \exp\left(-\frac{u^2}{2\Delta^2}\right).$$

Using Lemma D.1, we have for some absolute constant $C > 0$

$$\mathbb{E} \max_{\ell,\ell' \le m} |Y_{\ell,i} - \mathbb{E}Y_{\ell,\ell'}| \le C\Delta\sqrt{2\log m}. \tag{16}$$

Step 3: Now, we connect these auxiliary variables $Y_{\ell,\ell'}$ to regret definition. Fix $\ell \leq m$. Let the cluster contains the optimal action, i.e., $a^*(\theta) \in r_\ell$, we have

$$\begin{aligned}
\text{Regret} &= \max_{a \in \mathcal{A}_{\text{full}}} \mu_a - \max_{a' \in \mathcal{A}} \mu_{a'} \\
&= \max_{a \in r_\ell} \mu_a - \max_{\ell' \leq m} \mu_{a_{\ell'}} \\
&= \min_{\ell' \leq m} \left\{ \max_{a \in r_\ell} \mu_a - \mu_{a_{\ell'}} \right\} \\
&= \min_{\ell' \leq m} \left\{ \max_{a \in r_\ell \cup \{a_{\ell'}\}} \mu_a - \mu_{a_{\ell'}} \right\} \\
&= \min_{\ell' \leq m} Y_{\ell,\ell'} \\
&= \min_{\ell' \leq m} \max_{\ell'' \leq m} Y_{\ell'',\ell'}
\end{aligned}$$

The second equality uses $a^*(\theta) \in r_\ell$ and $\mathcal{A} = \{a_\ell\}_{\ell \leq m}$ is the reference subset. The fourth equality uses under the event of $a^*(\theta) \in r_\ell$, we have

$$\max_{a \in \mathcal{A}_{\text{full}}} \mu_a = \max_{a \in r_\ell} \mu_a \geq \mu_{a_{\ell'}}$$

such that $\max_{a \in r_\ell \cup \{a_{\ell'}\}} \mu_a = \max_{a \in r_\ell} \mu_a$. The fifth equality uses the definition of $Y_{\ell,\ell'}$ The last uses that under the event of $a^*(\theta) \in r_\ell$, we have $\max_{a \in r_\ell} \mu_a \geq \max_{a \in r_{\ell''}} \mu_a$ for any $\ell'' \leq m$, such that

$$Y_{\ell,\ell'} \geq Y_{\ell'',\ell'} \quad \forall \ell'' \leq m.$$

Therefore, since ℓ is included in $1, \ldots, m$, we reach the fact that

$$Y_{\ell,\ell'} = \max_{\ell'' \leq m} Y_{\ell'',\ell'}.$$

Note that this term $\min_{\ell' \leq m} \max_{\ell'' \leq m} Y_{\ell'',\ell'}$ is already irrelevant to ℓ. We conclude that for any θ,

$$\text{Regret} \geq \min_{\ell' \leq m} \max_{\ell \leq m} Y_{\ell,\ell'}.$$

Step 4: Using $Y_{\ell,\ell'} = \mathbb{E}Y_{\ell,\ell'} + (Y_{\ell,\ell'} - \mathbb{E}Y_{\ell,\ell'})$, we have

$$\begin{aligned}
\min_{\ell' \leq m} \max_{\ell \leq m} Y_{\ell,\ell'} &= \min_{\ell' \leq m} \max_{\ell \leq m} \Big(\mathbb{E}Y_{\ell,\ell'} + (Y_{\ell,\ell'} - \mathbb{E}Y_{\ell,\ell'}) \Big) \\
&\geq \min_{\ell' \leq m} \max_{\ell \leq m} \Big(\mathbb{E}Y_{\ell,\ell'} - |Y_{\ell,\ell'} - \mathbb{E}Y_{\ell,\ell'}| \Big) \\
&\geq \min_{\ell' \leq m} \max_{\ell \leq m} \Big(\mathbb{E}Y_{\ell,\ell'} - \max_{\ell'' \leq m} \max_{\ell''' \leq m} |Y_{\ell'',\ell'''} - \mathbb{E}Y_{\ell'',\ell'''}| \Big) \\
&= \min_{\ell' \leq m} \max_{\ell \leq m} \mathbb{E}Y_{\ell,\ell'} - \max_{\ell'' \leq m} \max_{\ell''' \leq m} |Y_{\ell'',\ell'''} - \mathbb{E}Y_{\ell'',\ell'''}|.
\end{aligned} \tag{17}$$

The first step uses the decomposition $Y_{\ell,\ell'} = \mathbb{E}Y_{\ell,\ell'} + (Y_{\ell,\ell'} - \mathbb{E}Y_{\ell,\ell'})$. The second step follows from the inequality $x \geq x - |x|$, applied pointwise. For the third step, we use that for all (ℓ, ℓ'),

$$|Y_{\ell,\ell'} - \mathbb{E}Y_{\ell,\ell'}| \leq \max_{\ell'' \leq m} \max_{\ell''' \leq m} |Y_{\ell'',\ell'''} - \mathbb{E}Y_{\ell'',\ell'''}|,$$

so each absolute value can be uniformly upper bounded. In the last step, the term

$$\max_{\ell'' \leq m} \max_{\ell''' \leq m} |Y_{\ell'',\ell'''} - \mathbb{E}Y_{\ell'',\ell'''}|$$

does not depend on ℓ, ℓ', so it can be taken outside the inner maximum.

Step 5: For every fixed pair (ℓ, ℓ'), one has

$$\min_{j \leq m} \mathbb{E}Y_{\ell,j} \;\leq\; \mathbb{E}Y_{\ell,\ell'} \;\leq\; \max_{i \leq m} \mathbb{E}Y_{i,\ell'}.$$

Since this holds for every ℓ, ℓ', it follows that for every ℓ, ℓ',

$$\min_{j \le m} \mathbb{E} Y_{\ell,j} \;\le\; \max_{i \le m} \mathbb{E} Y_{i,\ell'}.$$

Now taking $\max_{\ell \le m}$ on the left-hand side gives

$$\max_{\ell \le m} \min_{j \le m} \mathbb{E} Y_{\ell,j} \;\le\; \max_{i \le m} \mathbb{E} Y_{i,\ell'} \qquad \text{for every } \ell' \le m.$$

Finally, taking $\min_{\ell' \le m}$ on the right-hand side yields

$$\max_{\ell \le m} \min_{\ell' \le m} \mathbb{E} Y_{\ell,\ell'} \;\le\; \min_{\ell' \le m} \max_{\ell \le m} \mathbb{E} Y_{\ell,\ell'}.$$

Hence, we have

$$\min_{\ell' \le m} \max_{\ell \le m} \mathbb{E} Y_{\ell,\ell'} \;\ge\; \max_{\ell \le m} \min_{\ell' \le m} \mathbb{E} Y_{\ell,\ell'}. \tag{18}$$

Step 6: Recall the definition of $Y_{\ell,\ell'}$, fix ℓ

$$\min_{\ell' \le m} \mathbb{E} Y_{\ell,\ell'} = \min_{\ell' \le m} \mathbb{E}\left[\max_{a \in r_\ell \cup \{a_{\ell'}\}} \mu_a - \mu_{a_{\ell'}}\right] = \min_{\ell' \le m} \mathbb{E}\left[\max_{a \in r_\ell \cup \{a_{\ell'}\}} \mu_a\right] = \mathbb{E}\left[\max_{a \in r_\ell} \mu_a\right] = \mathbb{E}\left[\max_{a \in r_\ell} \mu_a - \mu_{a_\ell}\right], \tag{19}$$

where the first equality use the definition of $Y_{\ell,\ell'}$. The second uses $\mathbb{E}\mu_{a_{\ell'}} = 0$. The third uses $\max_{a \in r_\ell \cup \{a_{\ell'}\}} \mu_a \ge \max_{a \in r_\ell} \mu_a, \forall \ell' \le m$. The last uses again $\mathbb{E}\mu_{a_\ell} = 0$.

Step 7: Wrap up.

$$\begin{aligned}
\mathbb{E}[\text{Regret}] &\ge \min_{\ell' \le m} \max_{\ell \le m} \mathbb{E} Y_{\ell,\ell'} - \mathbb{E} \max_{\ell'' \le m} \max_{\ell''' \le m} |Y_{\ell'',\ell'''} - \mathbb{E} Y_{\ell'',\ell'''}| \\
&\ge \max_{\ell \le m} \min_{\ell' \le m} \mathbb{E} Y_{\ell,\ell'} - \mathbb{E} \max_{\ell'' \le m} \max_{\ell''' \le m} |Y_{\ell'',\ell'''} - \mathbb{E} Y_{\ell'',\ell'''}| \\
&\ge \max_{\ell \le m} \mathbb{E}\left[\max_{a \in r_\ell} \mu_a - \mu_{a_\ell}\right] - C\Delta\sqrt{2 \log m}.
\end{aligned}$$

The first use Equation (17). The second uses Equation (18). The last uses Equations (19) and (16). Let $C' = C\sqrt{2}$ completes the proof. □

F. Proof of Theorem 4.4

An important consequence of assuming $\mathbb{E}_\theta[\mu_a] = 0$ for all $a \in \mathcal{A}_{\text{full}}$ is the following property:

Lemma F.1 (Transformation invariance)**.** *Given any vector $x \in \mathbb{R}^n$, let $S + x := \{s + x : s \in \mathcal{S}\}$. If $\mathbb{E}[\theta] = 0$, then $\mathbb{E}[\max_{a \in \mathcal{S}+x} \langle a, \theta \rangle] = \mathbb{E}[\max_{a' \in \mathcal{S}} \langle a', \theta \rangle]$.*

Proof.

$$\begin{aligned}
\mathbb{E}\left[\max_{a \in \mathcal{S}+x} \langle a, \theta \rangle\right] &= \mathbb{E}\left[\max_{a' \in \mathcal{S}} \langle a' + x, \theta \rangle\right] \\
&= \mathbb{E}\left[\max_{a' \in \mathcal{S}} \langle a', \theta \rangle\right] + \mathbb{E}\left[\langle x, \theta \rangle\right] \\
&= \mathbb{E}\left[\max_{a' \in \mathcal{S}} \langle a', \theta \rangle\right],
\end{aligned}$$

where the last equality uses $\mathbb{E}[\langle x, \theta \rangle] = \langle x, \mathbb{E}[\theta] \rangle = 0$. □

The key tool for decomposing the regret of Algorithm 1 into that from a reference subset and that from missing clusters is the following Lemma:

Lemma F.2. *Consider a partition $\mathcal{R}$ of the full action space. Let $\mathcal{A}$ be the output of Algorithm 1. Then, the event that the optimal action falls in a cluster r, i.e., $\{a^*(\theta) \in r\}$, is independent of whether the subset $\mathcal{A}$ intersects with the cluster. It holds that*

$$\begin{aligned}\Pr[a^*(\theta) \in r,\, r \cap \mathcal{A} = \emptyset] =& q(r)(1-q(r))^K,\\ \Pr[a^*(\theta) \in r, r \cap \mathcal{A} \neq \emptyset] \leq& q(r).\end{aligned}$$

Proof. Since $\mathcal{A}$ is the output of Algorithm 1, the event $\{a^*(\theta) \in r\}$ is independent from $\{r \cap \mathcal{A} \neq \emptyset\}$ or $\{r \cap \mathcal{A} = \emptyset\}$. We have

$$\begin{aligned}\Pr[a^*(\theta) \in r, r \cap \mathcal{A} = \emptyset] =& \Pr[a^*(\theta) \in r] \Pr[r \cap \mathcal{A} = \emptyset]\\ =& q(r) \Pr[r \cap \mathcal{A} = \emptyset]\\ =& q(r)(1-q(r))^K,\end{aligned}$$

where the first equality uses independence between $\{a^*(\theta) \in r\}$ and $\{r \cap \mathcal{A} \neq \emptyset\}$, the second equality uses the definition of measure q, the third equality use the probability of missing cluster r in K i.i.d. samplings. Similarly,

$$\Pr[a^*(\theta) \in r, r \cap \mathcal{A} \neq \emptyset] = q(r) \cdot \Pr[r \cap \mathcal{A} \neq \emptyset].$$

Using $\Pr[r \cap \mathcal{A} \neq \emptyset] \leq 1$, we complete the proof. □

Statement of Theorem 4.4: Let $\mathcal{A}$ be output of Algorithm 1. Consider a partition $\mathcal{R} := \{r_\ell\}_{\ell \leq m}$ of the full action space, with $\epsilon := \max_{r \in \mathcal{R}} \operatorname{diam}(r)$. Then, for the same constant $C > 0$ in Theorem 4.2,

$$\begin{aligned}\mathbb{E}_{\theta,\mathcal{A}}[\text{Regret}] \leq& \max_{r \in \mathcal{R}} \mathbb{E}_\theta \left[\max_{a \in r} \mu_a\right] + C\epsilon\sqrt{\log m}\\ &+ \left(\mathbb{E}_q\left[(1-q(r))^{2K}\right] \cdot \mathbb{E}_\theta\left[\max_{a \in \mathcal{A}_{\text{full}}} \mu_a^2\right]\right)^{1/2}.\end{aligned}$$

Proof. If $r_\ell \cap \mathcal{A} \neq \emptyset$, define $a_\ell \in r_\ell \cap \mathcal{A}$. If $r_\ell \cap \mathcal{A} = \emptyset$, choose an arbitrary point $a_\ell \in r_\ell$ as the representative. The set $\mathcal{A}' := \{a_\ell\}_{\ell \leq m}$ forms a reference subset of Definition 4.1. The cluster r_ℓ is contained in a closed Euclidean ball of radius ϵ centered at a_ℓ, i.e., $r_\ell \subset B(a_\ell, \epsilon)$. Define a Gaussian process $\{Z_a\}_{a \in r_\ell}$ where $Z_a := \mu_a - \mu_{a_\ell}$. Define a random variable

$$Y_\ell := \sup_{a \in r_\ell} Z_a = \sup_{a \in r_\ell} \mu_a - \mu_{a_\ell}.$$

Since $\mathbb{E}\mu_a = 0$ for all $a \in \mathcal{A}_{\text{full}}$, we have $\mathbb{E}Y_\ell = \mathbb{E}\sup_{a \in r_\ell} \mu_a$.

Consider the case that $a^*(\theta) \in r_\ell$. We have

$$\begin{aligned}&\mathbb{E}\left[\text{Regret} \,\middle|\, r_\ell \cap \mathcal{A} \neq \emptyset, a^*(\theta) \in r_\ell\right]\\ \leq& \mathbb{E}\left[Y_\ell \,\middle|\, r_\ell \cap \mathcal{A} \neq \emptyset, a^*(\theta) \in r_\ell\right]\\ =& \mathbb{E}\left[Y_\ell \,\middle|\, a^*(\theta) \in r_\ell\right],\\ \leq& \mathbb{E}\left[\max_{\ell \leq m} Y_\ell \,\middle|\, a^*(\theta) \in r_\ell\right],\end{aligned} \tag{20}$$

where the first inequality uses $\text{Regret} \leq Y_\ell$ if $r_\ell \cap \mathcal{A} \neq \emptyset$, the equality uses that Y_ℓ is independent of $r_\ell \cap \mathcal{A} \neq \emptyset$. On the other hand, we assume that $0 \in \operatorname{Conv}(\mathcal{A})$ because even if it doesn't hold, we can always find a vector $x \in \mathbb{R}^n$ such that $0 \in \operatorname{Conv}(\mathcal{A} + x)$, without changing the value of $\mathbb{E}[\max_{a \in \mathcal{A}} \mu_a]$ (c.f. Lemma F.1). Therefore,

$$\begin{aligned}&\mathbb{E}\left[\text{Regret} \,\middle|\, r_\ell \cap \mathcal{A} = \emptyset, a^*(\theta) \in r_\ell\right]\\ \leq& \mathbb{E}\left[\max_{a \in \mathcal{A}_{\text{full}}} \mu_a \,\middle|\, a^*(\theta) \in r_\ell\right],\end{aligned} \tag{21}$$

where the inequality uses $\text{Regret} \leq \max_{a\in\mathcal{A}_{\text{full}}} \mu_a$, as a consequence of $0 \in \text{Conv}(\mathcal{A})$, and that $\max_{a\in\mathcal{A}_{\text{full}}} \mu_a$ is independent of $r_\ell \cap \mathcal{A} = \emptyset$. Further,

$$\begin{aligned}
\mathbb{E}[\text{Regret}] &= \sum_{r\in\mathcal{R}} \Pr[r\cap\mathcal{A}\neq\emptyset, a^*(\theta)\in r]\cdot\mathbb{E}\left[\text{Regret}\,\middle|\, r\cap\mathcal{A}\neq\emptyset, a^*(\theta)\in r\right] \\
&\quad + \sum_{r\in\mathcal{R}} \Pr[r\cap\mathcal{A}=\emptyset, a^*(\theta)\in r]\cdot\mathbb{E}\left[\text{Regret}\,\middle|\, r\cap\mathcal{A}=\emptyset, a^*(\theta)\in r\right] \\
&\leq \sum_{\ell\leq m} q(r_\ell)\cdot\left(\mathbb{E}\left[\max_{\ell\leq m} Y_\ell \,\middle|\, a^*(\theta)\in r_\ell\right] + (1-q(r_\ell))^K\cdot\mathbb{E}\left[\max_{a\in\mathcal{A}_{\text{full}}}\mu_a \,\middle|\, a^*(\theta)\in r_\ell\right]\right) \\
&= \mathbb{E}\left[\max_{\ell\leq m} Y_\ell\right] + \sum_{r\in\mathcal{R}} q(r)\cdot(1-q(r))^K\cdot\mathbb{E}\left[\max_{a\in\mathcal{A}_{\text{full}}}\mu_a \,\middle|\, a^*(\theta)\in r\right] \\
&\leq \mathbb{E}\left[\max_{\ell\leq m} Y_\ell\right] + \left(\mathbb{E}_q\left[(1-q(r))^{2K}\right]\cdot\mathbb{E}_\theta\left[\max_{a\in\mathcal{A}_{\text{full}}}\mu_a^2\right]\right)^{1/2},
\end{aligned} \tag{22}$$

where the first equality uses tower rule. The first inequality uses Equations (20-21), and Lemma F.2. The last equality uses tower rule again. The last inequality uses Cauchy-Schwarz inequality, which states that for any two random variables X and Y, we have $|\mathbb{E}[XY]| \leq \sqrt{\mathbb{E}[X^2]\mathbb{E}[Y^2]}$.

Note that the Gaussian process assumption is not used in (22); it is only needed to bound the term $\mathbb{E}\left[\max_{\ell\leq m} Y_\ell\right]$. By applying (15) in the proof of Theorem 4.2, we bound this term and complete the proof. □

G. Proof of Theorem 4.10

To analyze the worst-case behavior of Theorem 4.4, we first examine the term $\mathbb{E}_q\left[(1-q(r))^K\right]$ in the upper bound, summarized in the following Lemma:

Lemma G.1. *Let q denote a discrete probability distribution over a finite support $\mathcal{R}$. Define*

$$\begin{aligned}
M := \max_q\ & \mathbb{E}_q\left[(1-q(r))^K\right] \\
\text{s.t.}\ & \sum_{r\in\mathcal{R}} q(r) = 1,\ q(r)\in[0,1)\ \ \forall r\in\mathcal{R}.
\end{aligned} \tag{23}$$

When $m \geq K+1$, the maximum is $M = \left(1-\frac{1}{m}\right)^K$, and it is attained when q is uniform. When $m < K+1$, the maximum is upper bounded by $M \leq \frac{m}{K+1}\left(\frac{K}{K+1}\right)^K$, and there exists a feasible distribution q' such that $\mathbb{E}_{q'}\left[(1-q'(r))^K\right] \geq \frac{m-1}{K+1}\left(\frac{K}{K+1}\right)^K$.

Proof. Let $\mathcal{R} = \{r_1,\ldots,r_m\}$ be the finite support of the measure q, where $m := m$. Let $q_i := q(r_i)$ denote the probability mass at each support point. Define the function $f(q_\ell) := q_\ell\cdot(1-q_\ell)^K$.

Case I $m < K+1$**:** The first derivative of f is

$$f'(q_\ell) = (1-q_\ell)^{K-1}\big(1-(K+1)q_\ell\big),$$

which is positive on the interval $\left[0, \frac{1}{K+1}\right)$ and negative on the interval $\left(\frac{1}{K+1}, 1\right)$. Therefore, $f(q_\ell)$ attains its maximum over $[0,1)$ at

$$q_\ell^* = \frac{1}{K+1},$$

with the corresponding maximum value

$$f(q_\ell^*) \leq \frac{1}{K+1}\left(\frac{K}{K+1}\right)^K.$$

Since there are m support points, this yields the upper bound $M \leq \sum_{\ell=1}^{m} f(q_\ell^*) \leq \frac{m}{K+1}\left(\frac{K}{K+1}\right)^K$. Also, since $\frac{1}{K+1} < \frac{1}{m}$, the solution $q_1 = \cdots = q_{m-1} = \frac{1}{K+1}$ and $q_m = \frac{K-m+2}{K+1}$ is feasible. Thus,

$$\begin{aligned} M \geq & \sum_{\ell=1}^{m-1} f\left(\frac{1}{K+1}\right) + f\left(\frac{K-m+2}{K+1}\right) \\ \geq & \frac{m-1}{K+1}\left(\frac{K}{K+1}\right)^K, \end{aligned}$$

where the right inequality uses that $f(q_\ell) \geq 0$ for $q_\ell \in [0,1]$.

Case II $m \geq K+1$**:** Consider the relaxed maximization problem of (23):

$$\max_{q_1,\ldots,q_m \in [0,1)} \sum_{\ell=1}^{m} f(q_\ell) \quad \text{subject to} \sum_{\ell=1}^{m} q_\ell \leq 1. \tag{24}$$

Let $\lambda \geq 0$ be the Lagrange multiplier associated with the constraint. Define the Lagrangian:

$$\mathcal{L}(q_1,\ldots,q_m,\lambda) = \sum_{\ell=1}^{m} q_\ell(1-q_\ell)^K - \lambda\left(\sum_{\ell=1}^{m} q_\ell - 1\right).$$

For each $\ell = 1,\ldots,m$, compute the partial derivative of $\mathcal{L}$ with respect to q_ℓ:

$$\frac{\partial}{\partial q_\ell}\left[q_\ell(1-q_\ell)^K\right] = (1-q_\ell)^K - Kq_\ell(1-q_\ell)^{K-1}.$$

Setting this derivative equal to zero yields the stationary condition:

$$(1-q_\ell)^{K-1}(1-(K+1)q_\ell) = \lambda, \quad \text{with } \lambda \geq 0.$$

To find critical points of (24), we solve the system:

$$(1-q_\ell)^{K-1}(1-(K+1)q_\ell) = \lambda \geq 0 \quad \forall \ell \leq m, \quad \text{and} \quad \sum_{\ell=1}^{m} q_\ell \leq 1.$$

Define the function $g(q_\ell) := (1-q_\ell)^{K-1}(1-(K+1)q_\ell)$. For $g(q_\ell) \geq 0$, it must hold that $1-(K+1)q_\ell \geq 0$, i.e., $q_\ell \leq \frac{1}{K+1}$. Therefore, any feasible solution to this system must satisfy $q_\ell \in \left[0, \frac{1}{K+1}\right]$ for all $\ell \leq m$.

Then, over the interval $q_\ell \in \left[0, \frac{1}{K+1}\right]$, both factors $(1-q_\ell)^{K-1}$ and $1-(K+1)q_\ell$ are positive and decreasing. Hence, $g(q_\ell)$ is positive and strictly decreasing, so the equation $g(q_\ell) = \lambda \geq 0$ has at most one solution. Therefore, all q_ℓ's must be equal at a critical point. Let $q_\ell = c$ for all $\ell \leq m$. Due to the assumption of $m \geq K+1$, we have $\frac{1}{m} \leq \frac{1}{K+1}$, so any choice of $c \in \left[0, \frac{1}{m}\right]$ satisfies the constraint $\sum_{\ell=1}^{m} q_\ell \leq 1$ and is feasible for the system.

Therefore, $q_\ell = c$ for all $\ell \leq m$ is a feasible critical point of (24). The corresponding objective value is:

$$\sum_{\ell=1}^{m} f(c) = m \cdot c \cdot (1-c)^K,$$

which is increasing in c over the interval $\left[0, \frac{1}{m}\right]$. Hence, the maximum is attained at $c = \frac{1}{m}$, and the optimal value is:

$$\sum_{\ell=1}^{m} f\left(\frac{1}{m}\right) = \left(1-\frac{1}{m}\right)^K = \left(1-\frac{1}{m}\right)^K,$$

achieved when $q_\ell = \frac{1}{m}$ for all ℓ.

To confirm that this critical point is indeed a maximum, observe that the Hessian of the objective function $f(q_\ell)$ is diagonal (since the function is separable), and the diagonal entries are:

$$\frac{\partial^2}{\partial q_\ell^2}\left[q_\ell(1-q_\ell)^K\right] = -(1-q_\ell)^{K-2}\left(2K - K(K+1)q_\ell\right),$$

which is negative for $q_\ell \leq \frac{1}{K+1}$, because then $2-(K+1)q_\ell > 0$. Hence, the Hessian is negative definite, and the critical point is a local (and thus global) maximum.

Finally, note that (24) is a relaxation of (23). While the optimum of the original problem is upper bounded by that of the relaxed problem, the optimal solution to the relaxed problem also lies within the feasible region of the original problem. Therefore, the maximum of (23) is also attained when q is uniform, with the maximum value being $\left(1-\frac{1}{m}\right)^K$. □

Assumption G.2. The support of measure q is compact.

The compactness assumption ensures that the term $\mathbb{E}_\theta\left[\max_{a\in\mathcal{A}_{\text{full}}}\mu_a\right]$ is finite and guarantees the attainment of a unique optimal action. Without loss of generality, we assume that $\mathcal{A}_{\text{full}}$ is the support of the measure q.

Statement of Theorem 4.10 Under Assumption G.2, there exists a point a_0 and a constant $M>0$ such that $\mathcal{A}_{\text{full}} \subset B(a_0, M)$, a closed Euclidean ball of radius M centered at a_0. Let the action space have dimension $n \in \mathbb{N}$, and fix a constant $0<\epsilon<M$. Let $\mathcal{A}$ be the output of Algorithm 1. For the same constant $C>0$ in Theorem 4.2, we have:

$$\mathbb{E}_{\theta,\mathcal{A}}[\text{Regret}] \leq 2\epsilon\min\{\sqrt{n}, \sqrt{\frac{1}{2}\log|\mathcal{A}_{\text{full}}|+4}\} + C\epsilon\sqrt{\log N(\mathcal{A}_{\text{full}},\epsilon)}, \quad \text{where } K \geq \frac{1}{2}\left(\frac{M^2N(\mathcal{A}_{\text{full}},\epsilon)}{\epsilon^2 e}-1\right).$$

As $\epsilon \to 0^+$, we have $\mathbb{E}_{\theta,\mathcal{A}}[\text{Regret}] \to 0$ as $K\to\infty$.

Proof. Let $\{a_1,\dots,a_m\} \subseteq \mathcal{A}_{\text{full}}$ be a minimal geometric ϵ-net under the Euclidean norm, so that $m = N(\mathcal{A}_{\text{full}},\epsilon)$. Define $\pi(a)$ as the closest point in the ϵ-net to a, and let the partition $\mathcal{R} = \{r_1,\dots,r_m\}$ be given by

$$r_\ell = \{a \in \mathcal{A}_{\text{full}} : \pi(a) = a_\ell\}.$$

Step 1 bounding by dimension n:

By Lemma F.1, we can shift the action space to be centered at the origin. So, without loss of generality, we assume that $\mathcal{A}_{\text{full}} \subset M\cdot B_2^n$, the scaled unit Euclidean ball in $\mathbb{R}^n$. Then,

$$\mathbb{E}_\theta\left[\max_{a\in\mathcal{A}_{\text{full}}}\mu_a^2\right] \leq M^2\cdot\mathbb{E}\|\theta\|_2^2 = M^2 n, \tag{25}$$

where the inequality follows from $\mu_a = \langle\theta,a\rangle \leq M\|\theta\|_2$ for all $a\in MB_2^n$, and the equality uses that $\theta\sim\mathcal{N}(0,I)$, so each coordinate has variance 1 and $\|\theta\|_2^2 = \sum_{i=1}^n\theta_i^2$ has expectation n.

Then,

$$\max_{\ell\leq m}\mathbb{E}_\theta\left[\max_{a\in r_\ell}\mu_a\right] \leq \mathbb{E}_\theta\left[\max_{a\in B(a_\ell,\epsilon)}\mu_a\right] = \mathbb{E}_\theta\left[\max_{a\in\epsilon B_2^n}\mu_a\right] \leq \epsilon\sqrt{n}, \tag{26}$$

where the first inequality follows from $r_\ell \subset B(a_\ell,\epsilon)$ by construction; the equality uses Lemma F.1 to shift; and the final bound uses Property 3 from Supplementary J.

Step 2 bounding by cardinality $|\mathcal{A}_{\text{full}}|$:

By the definition of variance:

$$\begin{aligned}\mathbb{E}_\theta\left[\max_{a\in\mathcal{A}_{\text{full}}}\mu_a^2\right] &= \left(\mathbb{E}_\theta\left[\max_{a\in\mathcal{A}_{\text{full}}}\mu_a\right]\right)^2 + \text{Var}\left(\max_{a\in\mathcal{A}_{\text{full}}}\mu_a\right)\\ &\leq \left(\frac{\text{diam}(\mathcal{A}_{\text{full}})}{2}\sqrt{2\log|\mathcal{A}_{\text{full}}|}\right)^2 + 4\,\text{diam}(\mathcal{A}_{\text{full}})^2\\ &= M^2\left(\sqrt{\frac{1}{2}\log|\mathcal{A}_{\text{full}}|}+4\right)^2,\end{aligned} \tag{27}$$

where the inequality uses Supplementary K to bound the square of Gaussian width, and Lemma D.2 to bound the variance. The last equality uses the definition of M and simple rearrangement.

Also,

$$\max_{\ell \le m} \mathbb{E}_\theta \left[\max_{a \in r_\ell} \mu_a \right] \le \max_{\ell \le m} \frac{\mathrm{diam}(r_\ell)}{2} \sqrt{2 \log |r_\ell|} \le \frac{\epsilon}{2} \sqrt{2 \log |\mathcal{A}_{\text{full}}|}, \tag{28}$$

where the left inequality uses Supplementary K, the right inequality uses the definition of ϵ.

Then,

$$\begin{aligned}
\mathbb{E}_{\theta,\mathcal{A}}[\text{Regret}] \le & \max_{r \in \mathcal{R}} \mathbb{E}_\theta \left[\max_{a \in r} \mu_a \right] + C\epsilon\sqrt{\log m} + \left(\mathbb{E}_q \left[(1-q(r))^{2K} \right] \cdot \mathbb{E}_\theta \left[\max_{a \in \mathcal{A}_{\text{full}}} \mu_a^2 \right] \right)^{1/2}, \\
\le & \epsilon \min\{\sqrt{n}, \sqrt{\frac{1}{2}\log|\mathcal{A}_{\text{full}}|}\} + C\epsilon\sqrt{\log m} + \sqrt{\frac{m}{2K+1}} \left(\frac{2K}{2K+1} \right)^K M \ \min\{\sqrt{n}, \sqrt{\frac{1}{2}\log|\mathcal{A}_{\text{full}}|} + 4\} \\
\le & \epsilon \min\{\sqrt{n}, \sqrt{\frac{1}{2}\log|\mathcal{A}_{\text{full}}|}\} + C\epsilon\sqrt{\log m} + \epsilon \min\{\sqrt{n}, \sqrt{\frac{1}{2}\log|\mathcal{A}_{\text{full}}|} + 4\} \\
\le & 2\epsilon \min\{\sqrt{n}, \sqrt{\frac{1}{2}\log|\mathcal{A}_{\text{full}}|} + 4\} + C\epsilon\sqrt{\log N(\mathcal{A}_{\text{full}}, \epsilon)},
\end{aligned}$$

where the first inequality follows from the algorithm-dependent upper bound in Theorem 4.4. The second inequality follows from Equations (25)-(28), and the term $\mathbb{E}_q \left[(1-q(r))^{2K} \right]$ is bounded by $\frac{m}{2K+1} \left(\frac{2K}{2K+1} \right)^{2K}$ (see Lemma G.1). The third inequality comes from

$$\begin{aligned}
\log \left(\sqrt{\frac{m}{2K+1}} \left(\frac{2K}{2K+1} \right)^K M \right) = & \tfrac{1}{2} \log \left(\tfrac{m}{2K+1} \right) + K \log \left(\tfrac{2K}{2K+1} \right) + \log M \\
\le & \tfrac{1}{2} \log m - \tfrac{1}{2} \log(2K+1) - \tfrac{1}{2} + \log M \\
\le & \tfrac{1}{2} \log m - \tfrac{1}{2} \log \left(\frac{mM^2}{\epsilon^2 e} \right) - \tfrac{1}{2} + \log M \le \log \epsilon,
\end{aligned}$$

where the first inequality uses $\frac{1}{2} \log \left(\frac{m}{2K+1} \right) = \frac{1}{2}(\log m - \log(2K+1))$. Also, $\log \left(\frac{2K}{2K+1} \right) = \log \left(1 - \frac{1}{2K+1} \right) \le -\frac{1}{2K+1}$ due to the inequality $\log(1-X) \le -X$ for $X < 1$, and for a large K, we approximate $-\frac{K}{2K+1} \approx -\frac{1}{2}$. The last inequality uses the assumption $K \ge \frac{1}{2} \left(\frac{mM^2}{\epsilon^2 e} - 1 \right)$.

Finally, since $\mathcal{A}_{\text{full}} \subset MB_2^n$, the covering number satisfies $N(\mathcal{A}_{\text{full}}, \epsilon) \le C'(M/\epsilon)^n$ for some constant $C' > 0$; see Proposition 4.2.12 of (Vershynin, 2018). Since $\epsilon\sqrt{\log(M/\epsilon)} \to 0$ as $\epsilon \to 0^+$, it follows that $\epsilon\sqrt{\log N(\mathcal{A}_{\text{full}}, \epsilon)} \to 0$ as $\epsilon \to 0^+$. □

H. Proof of Theorem 4.11

For the any-algorithm lower bound, we derive a lower bound using techniques closely related to those used in the proof of Theorem 4.3. Given an arbitrary subset $\mathcal{A}$, with cardinality K, we define auxiliary variables $Y_{\ell,i}$ for $\ell \le m$ and $i \le K$.

Statement of Theorem 4.11. For any algorithm that outputs a subset $\mathcal{A}$ of cardinality K, for some absolute constant $C > 0$,

$$\mathbb{E}_\theta[\text{Regret}] \ \ge \ \max_{\ell \le m} \mathbb{E}_\theta \left[\max_{a \in r_\ell} \mu_a - \mu_{a_\ell} \right] - C \ \Delta(\mathcal{A}) \sqrt{\log m + \log K}.$$

Proof. **Step 1:** Fix a pair (ℓ, i), define a centered Gaussian process $\{Z_a\}_{a \in r_\ell \cup \{a_i\}}$ where $Z_a := \mu_a - \mu_{a_i}$. Define a non-negative random variable

$$Y_{\ell,i} := \sup_{a \in r_\ell \cup \{a_i\}} Z_a = \sup_{a \in r_\ell \cup \{a_i\}} \mu_a - \mu_{a_i}.$$

Step 2: Set $\Delta := \Delta(\mathcal{A})$. Further, by definition $\mathbb{E}Z_a^2 = \mathbb{E}(\mu_a - \mu_{a_i})^2 = \|a - a_i\|_2^2 \leq \Delta^2$. Using Lemma D.2 on the process $\{Z_a\}_{a \in r_\ell \cup \{a_i\}}$, we have

$$\Pr\left[|Y_{\ell,i} - \mathbb{E}Y_{\ell,i}| \geq u\right] \leq 2\exp\left(-\frac{u^2}{2\Delta^2}\right).$$

By union bound, we have

$$\Pr\left[\max_{\ell \leq m, i \leq K} |Y_{\ell,i} - \mathbb{E}Y_{\ell,i}| \geq u\right] \leq 2mK\exp\left(-\frac{u^2}{2\Delta^2}\right).$$

Using Lemma D.1, we have for some absolute constant $C > 0$

$$\mathbb{E}\max_{\ell \leq m, i \leq K} |Y_{\ell,i} - \mathbb{E}Y_{\ell,i}| \leq C\Delta\sqrt{\log mK}. \tag{29}$$

Step 3: Now, we connect these auxiliary variables $Y_{\ell,i}$ to regret definition. Fix $\ell \leq m$. Let the cluster contains the optimal action, i.e., $a^*(\theta) \in r_\ell$, we have

$$\begin{aligned}
\text{Regret} &= \max_{a \in \mathcal{A}_{\text{full}}} \mu_a - \max_{a' \in \mathcal{A}} \mu_{a'} \\
&= \max_{a \in r_\ell} \mu_a - \max_{i \leq K} \mu_{a_i} \\
&= \min_{i \leq K} \left\{\max_{a \in r_\ell} \mu_a - \mu_{a_i}\right\} \\
&= \min_{i \leq K} \left\{\max_{a \in r_\ell \cup \{a_i\}} \mu_a - \mu_{a_i}\right\} \\
&= \min_{i \leq K} Y_{\ell,i} \\
&= \min_{i \leq K} \max_{\ell' \leq m} Y_{\ell',i}
\end{aligned}$$

The second equality uses $a^*(\theta) \in r_\ell$ and $\mathcal{A} = \{a_i\}_{i \leq K}$ is action subset. The fourth equality uses under the event of $a^*(\theta) \in r_\ell$, we have

$$\max_{a \in \mathcal{A}_{\text{full}}} \mu_a = \max_{a \in r_\ell} \mu_a \geq \mu_{a_i}$$

such that $\max_{a \in r_\ell \cup \{a_i\}} \mu_a = \max_{a \in r_\ell} \mu_a$. The fifth equality uses the definition of $Y_{\ell,i}$ The last uses that under the event of $a^*(\theta) \in r_\ell$, we have $\max_{a \in r_\ell} \mu_a \geq \max_{a \in r_{\ell'}} \mu_a$ for any $\ell' \leq m$, such that

$$Y_{\ell,i} \geq Y_{\ell',\ell'} \quad \forall \ell' \leq m.$$

Therefore, since ℓ is included in $1, \ldots, m$, we reach the fact that

$$Y_{\ell,i} = \max_{\ell' \leq m} Y_{\ell',\ell'}.$$

Note that this term $\min_{i \leq K} \max_{\ell' \leq m} Y_{\ell',i}$ is already irrelevant to ℓ. We conclude that for any θ,

$$\text{Regret} \geq \min_{i \leq K} \max_{\ell \leq m} Y_{\ell,i}.$$

Step 4: Using $Y_{\ell,i} = \mathbb{E}Y_{\ell,i} + (Y_{\ell,i} - \mathbb{E}Y_{\ell,i})$, we have

$$\begin{aligned}
\min_{i \leq K} \max_{\ell \leq m} Y_{\ell,i} &= \min_{i \leq K} \max_{\ell \leq m} \Big(\mathbb{E}Y_{\ell,i} + (Y_{\ell,i} - \mathbb{E}Y_{\ell,i})\Big) \\
&\geq \min_{i \leq K} \max_{\ell \leq m} \Big(\mathbb{E}Y_{\ell,i} - |Y_{\ell,i} - \mathbb{E}Y_{\ell,i}|\Big) \\
&\geq \min_{i \leq K} \max_{\ell \leq m} \Big(\mathbb{E}Y_{\ell,i} - \max_{i' \leq K} \max_{\ell' \leq m} |Y_{\ell',i'} - \mathbb{E}Y_{\ell',i'}|\Big) \\
&= \min_{i \leq K} \max_{\ell \leq m} \mathbb{E}Y_{\ell,i} - \max_{i' \leq K} \max_{\ell' \leq m} |Y_{\ell',i'} - \mathbb{E}Y_{\ell',i'}|.
\end{aligned} \tag{30}$$

The first step uses the decomposition $Y_{\ell,i} = \mathbb{E}Y_{\ell,i} + (Y_{\ell,i} - \mathbb{E}Y_{\ell,i})$. The second step follows from the inequality $x \geq x - |x|$, applied pointwise. For the third step, we use that for all (ℓ, i),

$$|Y_{\ell,i} - \mathbb{E}Y_{\ell,i}| \leq \max_{i' \leq K} \max_{\ell' \leq m} |Y_{\ell',i'} - \mathbb{E}Y_{\ell',i'}|,$$

so each absolute value can be uniformly upper bounded. In the last step, the term

$$\max_{i' \leq K} \max_{\ell' \leq m} |Y_{\ell',i'} - \mathbb{E}Y_{\ell',i'}|$$

does not depend on (ℓ, i), so it can be taken outside the inner maximum.

Step 5: Recall the definition of $Y_{\ell,i}$, fix ℓ

$$\min_{i \leq K} \mathbb{E}Y_{\ell,i} = \min_{i \leq K} \mathbb{E}\left[\max_{a \in r_\ell \cup \{a_i\}} \mu_a - \mu_{a_i}\right] = \min_{i \leq K} \mathbb{E}\left[\max_{a \in r_\ell \cup \{a_i\}} \mu_a\right] \geq \mathbb{E}\left[\max_{a \in r_\ell} \mu_a\right] = \mathbb{E}\left[\max_{a \in r_\ell} \mu_a - \mu_{a_\ell}\right], \tag{31}$$

where the first equality use the definition of $Y_{\ell,i}$. The second uses $\mathbb{E}\mu_{a_i} = 0$. The inequality uses $\max_{a \in r_\ell \cup \{a_i\}} \mu_a \geq \max_{a \in r_\ell} \mu_a$ for all $i \leq K$. The last step uses again $\mathbb{E}\mu_{a_\ell} = 0$, where a_ℓ is the representative for cluster r_ℓ.

Step 7: Wrap up.

$$\begin{aligned}
\mathbb{E}[\text{Regret}] &\geq \min_{i \leq K} \max_{\ell \leq m} \mathbb{E}Y_{\ell,i} - \mathbb{E} \max_{i' \leq K} \max_{\ell' \leq m} |Y_{\ell',i'} - \mathbb{E}Y_{\ell',i'}| \\
&\geq \max_{\ell \leq m} \min_{i \leq K} \mathbb{E}Y_{\ell,i} - \mathbb{E} \max_{i' \leq K} \max_{\ell' \leq m} |Y_{\ell',i'} - \mathbb{E}Y_{\ell',i'}| \\
&\geq \max_{\ell \leq m} \mathbb{E}\left[\max_{a \in r_\ell} \mu_a - \mu_{a_i}\right] - C\Delta\sqrt{\log mK}.
\end{aligned}$$

The first use Equation (30). The second uses Equation (18). The last uses Equations (31) and (29). □

I. Proof of Theorem 4.13

The Gaussian process assumption can be relaxed to a sub-Gaussian one, where **the linear structure in Equation (5) is not required**, and the random process $\{\mu_a\}_{a \in \mathcal{S}}$ satisfies the following increment condition:

$$\forall u > 0,\ \Pr\left[|\mu_a - \mu_{a'}| \geq u\right] \leq 2\exp\left(-\frac{u^2}{2\|a - a'\|_2^2}\right). \tag{32}$$

We quantify the complexity of the action set using Talagrand's γ_2 functional (Talagrand, 2014).

Definition I.1. Let $T \subset \mathbb{R}^d$ be finite. The γ_2 functional is

$$\gamma_2(T) := \inf_{(T_n)} \sup_{t \in T} \sum_{n \geq 0} 2^{n/2} \operatorname{dist}(t, T_n),$$

where $(T_n)_{n \geq 0}$ is a sequence of subsets of T. The infimum is taken over all subsets of T such that

- $T_0 = \{t_0\}$,
- $|T_n| \leq 2^{2^n}$ for all $n \geq 1$,
- $\operatorname{dist}(t, T_n) := \min_{s \in T_n} \|t - s\|_2$.

Using this notion of complexity, we can establish the regret bound, based on the following lemma.

Lemma I.2. *Let $T \subset \mathbb{R}^d$ be finite. For every $u \geq 1$,*

$$\mathbb{P}\left(\sup_{s,t \in T} |X_t - X_s| \leq 8\,\gamma_2(T) + 16\,u\ \operatorname{diam}(T)\right) \geq 1 - 2e^{-u^2}.$$

Proof. Since T is finite, there exists an admissible sequence $(T_n)_{n\geq 0}$ such that

$$|T_0| = 1, \qquad |T_n| \leq 2^{2^n}, \qquad \sup_{t\in T} \sum_{n\geq 0} 2^{n/2} \mathrm{dist}(t, T_n) = \gamma_2(T),$$

where $\mathrm{dist}(t, T_n) := \inf_{a\in T_n} \|t - a\|_2$.

For each $t \in T$, choose $\pi_n(t) \in T_n$ such that

$$\|t - \pi_n(t)\|_2 = \mathrm{dist}(t, T_n).$$

For n large enough we have $T_n = T$, hence $\pi_n(t) = t$.

Fix $t_0 \in T_0$. Then

$$\sup_{s,t\in T} |X_t - X_s| \leq 2 \sup_{t\in T} |X_t - X_{t_0}|.$$

Fix $u \geq 1$ and choose κ such that

$$2^{\kappa/2} \leq u < 2^{(\kappa+1)/2}.$$

For each $t \in T$,

$$X_t - X_{t_0} = (X_{\pi_\kappa(t)} - X_{t_0}) + \sum_{n>\kappa} \big(X_{\pi_n(t)} - X_{\pi_{n-1}(t)}\big).$$

Step 1: Coarse part.

Since $\pi_\kappa(t) \in T_\kappa$,

$$\sup_{t\in T} |X_{\pi_\kappa(t)} - X_{t_0}| \leq \max_{a\in T_\kappa} |X_a - X_{t_0}|.$$

For each $a \in T_\kappa$,

$$\mathbb{P}\big(|X_a - X_{t_0}| > r \ \mathrm{diam}(T)\big) \leq 2e^{-r^2/2}.$$

Using $|T_\kappa| \leq 2^{2^\kappa}$ and union bound,

$$\mathbb{P}\left(\max_{a\in T_\kappa} |X_a - X_{t_0}| > r \ \mathrm{diam}(T)\right) \leq 2^{2^\kappa} \cdot 2e^{-r^2/2}.$$

Take $r = 4u$. Since $2^\kappa \leq u^2$,

$$2^{2^\kappa} \cdot 2e^{-8u^2} \leq 2\exp\big((\log 2)u^2 - 8u^2\big) \leq e^{-u^2}.$$

Thus with probability at least $1 - e^{-u^2}$,

$$\sup_{t\in T} |X_{\pi_\kappa(t)} - X_{t_0}| \leq 4u \ \mathrm{diam}(T).$$

Step 2: Fine part.

For $n > \kappa$, define the event

$$E_n := \left\{|X_{\pi_n(t)} - X_{\pi_{n-1}(t)}| \leq 4 \cdot 2^{n/2} \|\pi_n(t) - \pi_{n-1}(t)\|_2 \text{ for all } t \in T\right\}.$$

There are at most

$$|T_n||T_{n-1}| \leq 2^{2^n} 2^{2^{n-1}} \leq 2^{2^{n+1}}$$

pairs (a, b) to check. For each,

$$\mathbb{P}\big(|X_a - X_b| > 4 \cdot 2^{n/2} \|a - b\|_2\big) \leq 2e^{-8\cdot 2^n/2} = 2e^{-4\cdot 2^n}.$$

Hence

$$\mathbb{P}(E_n^c) \le 2^{2^{n+1}} \cdot 2e^{-4\cdot 2^n} \le 2e^{-2^n}.$$

Therefore

$$\mathbb{P}\Big(\bigcup_{n>\kappa} E_n^c\Big) \le \sum_{n>\kappa} 2e^{-2^n} \le e^{-u^2}.$$

On the event $\cap_{n>\kappa} E_n$, for every $t \in T$,

$$\sum_{n>\kappa} |X_{\pi_n(t)} - X_{\pi_{n-1}(t)}| \le 4 \sum_{n>\kappa} 2^{n/2} \|\pi_n(t) - \pi_{n-1}(t)\|_2.$$

Using

$$\|\pi_n(t) - \pi_{n-1}(t)\|_2 \le \mathrm{dist}(t, T_n) + d(t, T_{n-1}),$$

we obtain

$$\sum_{n>\kappa} 2^{n/2} \|\pi_n(t) - \pi_{n-1}(t)\|_2 \le \sum_{n>\kappa} 2^{n/2} \mathrm{dist}(t, T_n) + \sum_{n>\kappa} 2^{n/2} d(t, T_{n-1}).$$

The second term satisfies

$$\sum_{n>\kappa} 2^{n/2} d(t, T_{n-1}) = \sqrt{2} \sum_{k\ge\kappa} 2^{k/2} d(t, T_k).$$

Hence

$$\sum_{n>\kappa} 2^{n/2} \|\pi_n(t) - \pi_{n-1}(t)\|_2 \le (1+\sqrt{2}) \sum_{k\ge 0} 2^{k/2} d(t, T_k).$$

Taking supremum over t,

$$\sup_{t\in T} \sum_{n>\kappa} |X_{\pi_n(t)} - X_{\pi_{n-1}(t)}| \le 4(1+\sqrt{2})\, \gamma_2(T).$$

Step 3: Wrap up.

With probability at least $1 - 2e^{-u^2}$, both bounds hold, and therefore

$$\sup_{t\in T} |X_t - X_{t_0}| \le 4(1+\sqrt{2})\, \gamma_2(T) + 4u \ \mathrm{diam}(T).$$

Thus

$$\sup_{s,t\in T} |X_t - X_s| \le 2 \sup_{t\in T} |X_t - X_{t_0}| \le 8(1+\sqrt{2})\, \gamma_2(T) + 8u \ \mathrm{diam}(T).$$

Since $1 + \sqrt{2} < 3$, we may simplify:

$$\sup_{s,t\in T} |X_t - X_s| \le 8\, \gamma_2(T) + 16\, u \ \mathrm{diam}(T).$$

This completes the proof. □

Statement of Theorem 4.13: Let $\mathcal{A}$ be output of Algorithm 1. Let $\{\mu_a\}_{a\in\mathcal{A}_{\text{full}}}$ be a mean-zero sub-Gaussian process satisfying Equation (32). Consider a partition $\mathcal{R} := \{r_\ell\}_{\ell\le m}$ of the full action space, where $m \ge 2$. Then,

$$\mathbb{E}_{\theta,\mathcal{A}}[\text{Regret}] \le 8 \max_{\ell\le m} \gamma_2(r_\ell) + 57\, \epsilon \sqrt{\log(m)} + \left(\mathbb{E}_q\left[(1-q(r))^{2K}\right] \cdot \mathbb{E}_\theta\left[\max_{a\in\mathcal{A}_{\text{full}}} \mu_a^2\right]\right)^{1/2}.$$

Proof. **Step 1: Auxiliary variables.** If $r_\ell \cap \mathcal{A} \neq \emptyset$, define $a_\ell \in r_\ell \cap \mathcal{A}$. If $r_\ell \cap \mathcal{A} = \emptyset$, choose an arbitrary point $a_\ell \in r_\ell$ as the representative. The set $\mathcal{A}' := \{a_\ell\}_{\ell \leq m}$ forms a reference subset of Definition 4.1.

For each $\ell \leq m$, define a random variable

$$Y_\ell := \sup_{a \in r_\ell} \mu_a - \mu_{a_\ell}.$$

Step 2: By Lemma I.2, we have for every $u \geq 1$,

$$\mathbb{P}\left(\sup_{s,t \in r_\ell} |\mu_t - \mu_s| \leq 8\,\gamma_2(r_\ell) + 16\,u\ \mathrm{diam}(r_\ell) \right) \geq 1 - 2e^{-u^2}.$$

A simple rearrangement gives for every $\nu \geq 0$,

$$\mathbb{P}\left(\sup_{s,t \in r_\ell} |\mu_t - \mu_s| > 8\,\gamma_2(r_\ell) + 16\ \mathrm{diam}(r_\ell) + \nu \right) \leq 2\exp\left(-\frac{\nu^2}{16^2\,\mathrm{diam}(r_\ell)^2} \right).$$

Recall that

$$Y_\ell = \sup_{a \in r_\ell} \mu_a - \mu_{a_\ell} \ \leq \ \sup_{s,t \in r_\ell} |\mu_t - \mu_s|,$$

and define

$$\epsilon := \max_{\ell \leq m} \mathrm{diam}(r_\ell).$$

Moreover, set

$$M := \max_{\ell \leq m}\ 8\,\gamma_2(r_\ell) + 16\,\epsilon.$$

By a union bound,

$$\mathbb{P}\left(\max_{\ell \leq m} Y_\ell > M + \nu \right) \leq 2m\exp\left(-\frac{\nu^2}{16^2\epsilon^2} \right). \tag{33}$$

Step 3:

For any $\nu \geq 0$, using Equation (33),

$$\begin{aligned} \mathbb{E}\max_{\ell \leq m} Y_\ell &\leq M + \nu + \int_\nu^\infty \mathbb{P}\left(\max_{\ell \leq m} Y_\ell > M + t \right)\, dt \\ &\leq M + \nu + \int_\nu^\infty 2m\exp\left(-\frac{t^2}{16^2\epsilon^2} \right)\, dt. \end{aligned}$$

Choose

$$\nu = 16\,\epsilon\sqrt{\log(2m)}.$$

Then for all $t \geq \nu$,

$$2m\exp\left(-\frac{t^2}{16^2\epsilon^2} \right) = \exp\left(-\frac{t^2 - \nu^2}{16^2\epsilon^2} \right).$$

Using the Gaussian tail bound

$$\int_\nu^\infty e^{-t^2/a^2}\, dt \leq \frac{a^2}{2\nu} e^{-\nu^2/a^2}, \qquad \text{where } a = 16\epsilon,$$

we obtain

$$\int_\nu^\infty 2m\exp\left(-\frac{t^2}{16^2\epsilon^2} \right)\, dt \leq \frac{16^2\epsilon^2}{2\nu}.$$

Therefore, for $m \geq 2$,

$$\begin{aligned}\mathbb{E}\max_{\ell\leq m} Y_\ell \leq& M + 16\,\epsilon\sqrt{\log(2m)} + \frac{8\,\epsilon}{\sqrt{\log(2m)}}\\ \leq& M + 24\,\epsilon\sqrt{\log(2m)}\\ =& \max_{\ell\leq m} 8\,\gamma_2(r_\ell) + 16\,\epsilon + 24\,\epsilon\sqrt{\log(2m)}\\ \leq& 8\max_{\ell\leq m}\gamma_2(r_\ell) + +57\,\epsilon\sqrt{\log(m)}.\end{aligned} \tag{34}$$

Step 4: Following the same reasoning used in the proof of Theorem 4.4:

$$\begin{aligned}\mathbb{E}[\text{Regret}] \leq& \mathbb{E}\left[\max_{\ell\leq m} Y_\ell\right] + \left(\mathbb{E}_q\left[(1-q(r))^{2K}\right]\cdot\mathbb{E}_\theta\left[\max_{a\in\mathcal{A}_{\text{full}}}\mu_a^2\right]\right)^{1/2}\\ \leq& 8\max_{\ell\leq m}\gamma_2(r_\ell) + +57\,\epsilon\sqrt{\log(m)} + \left(\mathbb{E}_q\left[(1-q(r))^{2K}\right]\cdot\mathbb{E}_\theta\left[\max_{a\in\mathcal{A}_{\text{full}}}\mu_a^2\right]\right)^{1/2},\end{aligned}$$

where the first inequality follows from (22) in the proof of Theorem 4.4. The last inequality uses Equation (34). □

J. Properties of Gaussian width

Given a set $\mathcal{S} \in \mathbb{R}^n$, the term $\mathbb{E}[\max_{a\in\mathcal{S}}\mu_a]$ where $\theta \sim \mathcal{N}(0, I)$ is called Gaussian (mean) width.

Property 1: $\mathbb{E}\left[\max_{a\in\mathcal{S}}\langle a,\theta\rangle\right] = \mathbb{E}\left[\max_{a'\in-\mathcal{S}}\langle a',\theta\rangle\right]$**.**

$$\mathbb{E}\left[\max_{a\in\mathcal{S}}\langle a,\theta\rangle\right] = \mathbb{E}\left[\max_{a\in\mathcal{S}}\langle a,-\theta\rangle\right] = \mathbb{E}\left[\max_{a\in\mathcal{S}}\langle -a,\theta\rangle\right] = \mathbb{E}\left[\max_{a'\in-\mathcal{S}}\langle a',\theta\rangle\right],$$

where the first equality uses $-\theta$ and θ are identically distributed. The third equality uses for any $a \in \mathcal{S}$, it holds that $-a \in -\mathcal{S}$.

Property 2: $\mathbb{E}\left[\max_{a\in\mathcal{S}}\langle a,\theta\rangle\right] \leq \frac{1}{2}\mathbb{E}\left[\max_{a,a'\in\mathcal{S}}\langle a-a',\theta\rangle\right]$**.** Let $a^*(-\mathcal{S},\theta)$ denote the optimal action in $\mathcal{S}$ for bandit instance θ.

$$\begin{aligned}2\cdot\mathbb{E}\left[\max_{a\in\mathcal{S}}\langle a,\theta\rangle\right] =& \mathbb{E}\left[\max_{a\in\mathcal{S}}\langle a,\theta\rangle\right] + \mathbb{E}\left[\max_{a'\in-\mathcal{S}}\langle a',\theta\rangle\right]\\ =& \mathbb{E}\left[\langle a^*(\mathcal{S},\theta),\ \theta\rangle\right] + \mathbb{E}\left[\langle a^*(-\mathcal{S},\theta),\ \theta\rangle\right]\\ =& \mathbb{E}\left[\langle a^*(\mathcal{S},\theta) + a^*(-\mathcal{S},\theta),\ \theta\rangle\right]\\ \leq& \mathbb{E}\left[\max_{a,a'\in\mathcal{S}}\langle a-a',\theta\rangle\right],\end{aligned}$$

where the first equality uses Property 1, the second equality uses the definition of $a^*(-\mathcal{S},\theta)$. The third equality uses linearity of expectation. The inequality uses that the vector $a^*(\mathcal{S},\theta) + a^*(-\mathcal{S},\theta)$ belongs to the set of vectors $\{a - a' : a, a' \in \mathcal{S}\}$. In fact, one can prove the equality that $\mathbb{E}\left[\max_{a\in\mathcal{S}}\langle a,\theta\rangle\right] = \frac{1}{2}\mathbb{E}\left[\max_{a,a'\in\mathcal{S}}\langle a-a',\theta\rangle\right]$, but we only need inequality to prove the next claim.

Property 3: $\mathbb{E}\left[\max_{a\in\mathcal{S}}\mu_a\right] \leq \frac{\text{diam}(\mathcal{S})}{2}\cdot\sqrt{n}$**.**

$$\begin{aligned}\mathbb{E}\left[\max_{a\in\mathcal{S}}\mu_a\right] =& \mathbb{E}\left[\max_{a\in\mathcal{S}}\langle a,\theta\rangle\right]\\ \leq& \frac{1}{2}\mathbb{E}\left[\max_{a,a'\in\mathcal{S}}\langle a-a',\theta\rangle\right]\\ \leq& \frac{1}{2}\mathbb{E}\max_{a,a'\in\mathcal{S}}\|\theta\|_2\|a-a'\|_2\\ \leq& \frac{1}{2}\mathbb{E}\,\text{diam}(\mathcal{S})\|\theta\|_2 \leq \frac{\text{diam}(\mathcal{S})}{2}\cdot\sqrt{n},\end{aligned}$$

where the first equality uses the definition of μ_a. The first inequality uses Property 2. The second inequality uses Cauchy-Schwarz inequality. The third inequality uses the definition of $\mathrm{diam}(\cdot)$. The last inequality uses $\mathbb{E}\|\theta\|_2 \leq \sqrt{n}$.

K. Bounds of expected maximum of Gaussian

Let $X_1, \ldots, X_N$ be N random Gaussian variables (no necessarily independent) with zero mean and variance of marginals smaller than σ^2, then

$$\mathbb{E}\left[\max_{i=1,\ldots,N} X_i\right] \leq \sigma\sqrt{2\log N}.$$

Proof. for any $\delta > 0$,

$$\begin{aligned}\mathbb{E}\left[\max_{i=1,\ldots,N} X_i\right] =& \frac{1}{\delta}\mathbb{E}\left[\log\exp(\delta \max_{i=1,\ldots,N} X_i)\right] \leq \frac{1}{\delta}\log\mathbb{E}\left[\exp(\delta \max_{i=1,\ldots,N} X_i)\right] \\ =& \frac{1}{\delta}\log\mathbb{E}\left[\max_{i=1,\ldots,N}\exp(\delta X_i)\right] \leq \frac{1}{\delta}\log\sum_{i=1}^{N}\mathbb{E}\left[\exp(\delta X_i)\right] \\ \leq& \frac{1}{\delta}\log\sum_{i=1}^{N}\exp(\sigma^2\delta^2/2) = \frac{\log N}{\delta} + \frac{\sigma^2\delta}{2},\end{aligned}$$

where the first inequality uses Jensen's inequality. Taking $\delta := \sqrt{2(\log N)/\sigma^2}$ yields the results. □

Let $X_1, \ldots, X_N$ be i.i.d. $\mathcal{N}(0, \sigma^2)$ random variables, then according to (Kamath, 2015):

$$\mathbb{E}_\theta\left[\max_{i=1,\ldots,N} X_i\right] \geq \frac{\sigma\sqrt{\log N}}{\sqrt{\pi\log 2}}.$$

L. The effect of clustering structure on regret

In Figure 4, we study the effect of the cluster diameters (controlled by a spread parameter) on regret. We structure the clustered action space: Five center points are fixed on the unit sphere in $\mathbb{R}^3$, and around each center, 200 points are sampled to form five clusters. Each point is obtained by adding Gaussian noise (mean zero, standard deviation equal to the spread parameter) to the center direction, followed by projection back onto the unit sphere. Bandits are sampled from a 3-dimensional standard Gaussian distribution, i.e., $\theta \sim \mathcal{N}(0, I)$. The left subplot shows the expected regret of Algorithm 1 with $K = 10$, computed using 10^4 additional bandits, as the spread varies from 0.01 to 0.5. The curves and error shade represent the mean $\pm$ one standard deviation of expected regret over 30 repetitions. The middle and right subplots display example action spaces for spread values of 0.01 and 0.5, with representative actions (purple stars) selected by Algorithm 1 with $K = 10$.

M. Varying-dependence actions with RBF/Gibbs kernels

We study the effect of varying action dependence using a Gaussian process with a kernel. To control the degree of dependence, we use stationary RBF kernel and non-stationary Gibbs kernel (Williams & Rasmussen, 2006).

$$\begin{aligned}\underset{\text{RBF}}{k}(a, a') =& \exp\left(-\frac{\|a - a'\|^2}{2l^2}\right), \\ \underset{\text{Gibbs}}{k}(a, a') =& \sqrt{\frac{2\,l(a)l(a')}{l(a)^2 + l(a')^2}}\exp\left(-\frac{\|a - a'\|^2}{l(a)^2 + l(a')^2}\right),\end{aligned} \tag{35}$$

where l is a length-scale parameter and $l(a) := 0.1 + 0.9 \cdot \exp(-\|a\|^2)$ is a location-dependent length-scale function. Both of them control the dependence between actions. But, unlike the stationary RBF kernels, the Gibbs kernel allows the correlation to depend not only on the distance between actions, but also on their locations. When $l(a) = l$ is a constant, the Gibbs kernel reduces to the RBF kernel.

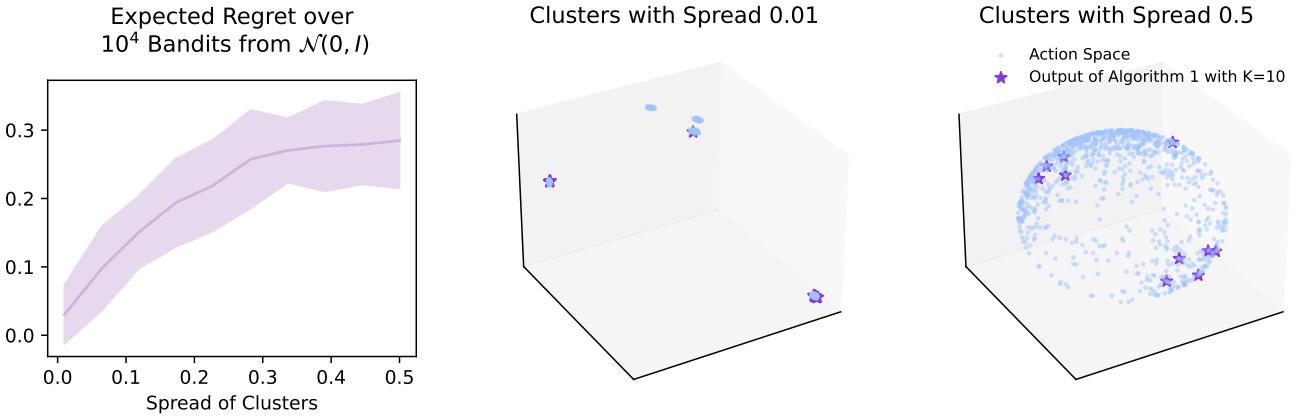

Figure 4. Illustration of clustered action spaces on unit sphere in $\mathbb{R}^3$ and the effect of cluster diameters on regret. Five clusters are formed by generating 5 fixed center points, with 200 points sampled around each using Gaussian noise (spread controls the variance). Bandits are drawn from $\mathcal{N}(0, I)$. The left subplot shows the mean $\pm$ standard deviation of the expected regret (over 30 trials) as the spread varies from 0.01 to 0.5, using 10^4 additional bandits. The middle and right subplots show example action spaces (blue dots) for spread values 0.01 and 0.5, with representative actions (purple stars) selected by Algorithm 1 with $K = 10$.

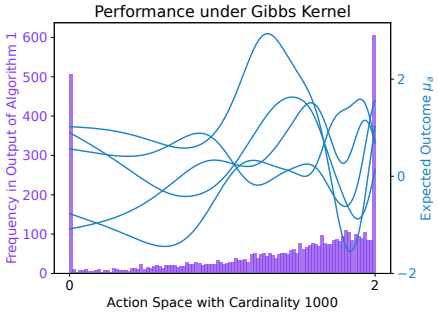

Figure 5. Experiments with outcome functions sampled from RBF/Gibbs kernels in (35). Sampled outcome functions from Gibbs kernel over fixed 1000 grid points in $[0, 2]$ (blue curves, right y-axis). The histogram (purple bars, left y-axis) shows action selection frequencies by Algorithm 1 with $K = 5000$, favoring regions with rougher functions and edge points.

Sampling Outcome Functions from a Kernel: We first construct the kernel matrix $\mathbf{K}$, where each entry is given by $\mathbf{K}_{a,a'} = k(a, a')$, for $a, a' \in \mathcal{A}_{\text{full}}$, depending on the choice of kernel. We then sample a Gaussian vector (a Gaussian process function evaluated at finite input) $f \sim \mathcal{N}(0, \mathbf{K})$. Under either kernels defined in (35), the variance of the function value $f(a)$ is one for all $a \in \mathcal{A}_{\text{full}}$. In this way, we sample functions from a RKHS function class; See (Kanagawa et al., 2018, Theorem 4.12).

To study the effect of varying dependence on the output of Algorithm 1, we consider a fixed action space consisting of 1000 grid points in the interval $[0, 2]$, using Gibbs kernel in (35) to sample outcome functions. To simplify computations, we marginalize a Gaussian process defined by the kernel over the grid. Figure 5 provides examples of sampled outcome functions (blue curves, with the y-axis on the right-hand side), which become smoother as the actions approach the left end of the interval—indicating stronger correlations among function values in that region. We run Algorithm 1 on this action space with $K = 5000$ to select actions and record the frequency of each action being selected. The resulting histogram (purple bars, with the y-axis on the left) reflects the importance measure q, highlighting that Algorithm 1 tends to select more actions from regions where the outcome functions are rougher—i.e., where action outcomes are less correlated and their features $\Phi(a)$ are are farther apart. Another interesting aspect of this subplot is the two high bars at the edges. Recall that the actual action space consists of feature vectors $\Phi(a)$ for $a \in \mathcal{A}_{ful}$. For actions indexed closer to 0, their feature vectors become more densely packed compared to those indexed closer to 2, resulting in more correlated outcomes. The two actions at the edges, indexed by 0 and 2, correspond to the two farthest points in the actual feature space.

N. Independent and identically distributed actions

As an extreme case, suppose that $\{\mu_a\}_{a\in\mathcal{A}_{\text{full}}}$ is a set of i.i.d. random variables. This corresponds to the canonical process in which the action space is given by the orthonormal basis of $\mathbb{R}^n$, where $n = |\mathcal{A}_{\text{full}}|$. To see this, we associate each action a with a unit vector e_a, which has a value of 1 at the ath coordinate and 0 elsewhere, and define the expected outcome as $\mu_a(\theta) := \langle e_a, \theta\rangle$. With this construction, the collection $\{\mu_a\}_{a\in\mathcal{A}_{\text{full}}}$ consists of mutually independent random variables.

Let the set $\mathcal{A}_{\text{full}} = \{e_i : i = 1, \dots, n\}$ denote the n unit vectors aligned with the coordinate axes in $\mathbb{R}^n$. Hence, $\text{diam}(\mathcal{A}_{\text{full}}) = \sqrt{2}$. In this case, $\max_{a\in\mathcal{A}_{\text{full}}}\langle a, \theta\rangle$ is equivalent to the maximum among the n entries of θ, where each entry is i.i.d. from the standard normal distribution $\mathcal{N}(0, 1)$. By symmetry, each coordinate has the same probability of attaining the maximum value. If each cluster contains only one unit vector, then the importance measure q over clusters is the uniform distribution.

Expected maximum in $\mathcal{A}_{\text{full}}$: Let X_i, $i = 1, \dots, n$ be i.i.d. samples from $\mathcal{N}(0, 1)$, then

$$\mathbb{E}\max_{a\in\mathcal{A}_{\text{full}}}\mu_a = \mathbb{E}_\theta\max_{i=1,\dots,n}\theta_i = \mathbb{E}\max_{i=1,\dots,n}X_i,$$

where the equivalence comes from that each entry θ_i is i.i.d. sample from $\mathcal{N}(0, 1)$. Then, according to the bounds of expected maximum of Gaussian in Supplementary K, we have

$$\frac{\sqrt{\log n}}{\sqrt{\pi\log 2}} \le \mathbb{E}\max_{a\in\mathcal{A}_{\text{full}}}\mu_a \le \sqrt{2\log n}, \quad \frac{\sqrt{\log|\mathcal{A}|}}{\sqrt{\pi\log 2}} \le \mathbb{E}\max_{a\in\mathcal{A}}\mu_a \le \sqrt{2\log|\mathcal{A}|}.$$

Bounds of expected regret of arbitrary $\mathcal{A}$: By definition of regret in (1),

$$\frac{\sqrt{\log n}}{\sqrt{\pi\log 2}} - \sqrt{2\log|\mathcal{A}|} \le \mathbb{E}_\theta[\text{Regret}] \le \sqrt{2\log n} - \frac{\sqrt{\log|\mathcal{A}|}}{\sqrt{\pi\log 2}}.$$

As a result, any algorithm, including Algorithm 1, would perform poorly unless the subset size $|\mathcal{A}|$ is sufficiently large.